



MATHS in Technology



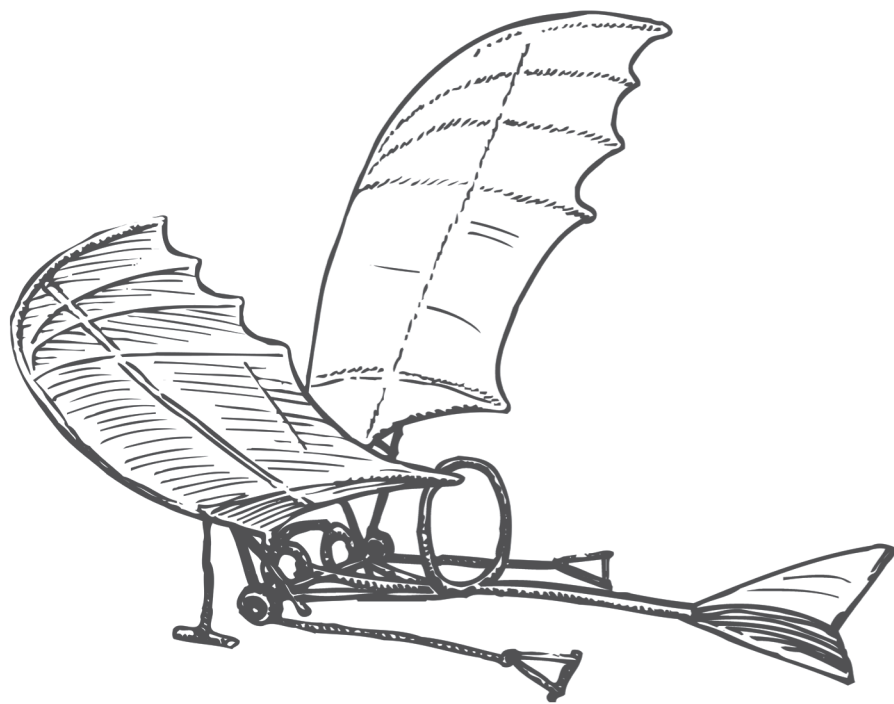
Barbara Lynch



Quality ready-to-use
resources



MATHS in Technology



Barbara Lynch



Barbara Lynch is former Head of Wesley College in Glen Waverley, Victoria, author of over 80 titles and now a consultant in areas such as strategic planning, governance, policy development, curriculum, assessment, reporting, CD-ROM and Internet publishing. Barbara has a long and enviable background in secondary education and mathematics.

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Maths in Technology is a collection of reproducible student worksheets that feature the application of mathematical knowledge and techniques in various areas of Technology. Any Technology theory required is provided as part of the worksheet. Students are given an 'Idea' of the answer in some cases.

Technology is the application of scientific advances to areas that will benefit people and includes any tools that help us to improve the way we can do or think about things. Hence technology can range from a pencil and paper to the latest gadget, from engines to satellites, from an abacus to a computer.

The series has been designed to cover Years 9 and 10 syllabi but the worksheets could be used to motivate and add interest to any Junior or Middle school Mathematics course. In most cases, an electronic calculator would be useful.

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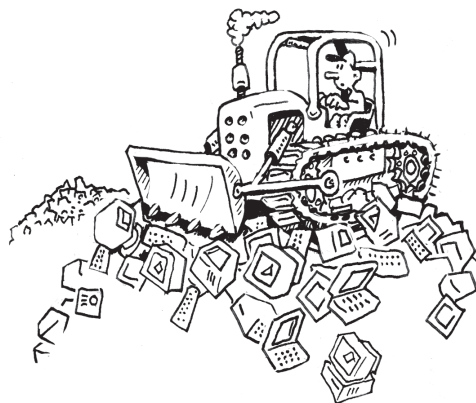
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Technology facts and figures



1

The adoption of technology has been rapid and extensive. Whether it be the Internet, mobile phones, computers, radios or televisions, people in the developed countries have become dependent on their 'gadgets'.



- 1 Approximately how many people used the Internet:

a in 1995?

Idea: 20 followed by $\frac{2 \times 8 \times 3 \times 5}{4 \times 10}$ zeros

b in 2000?

Idea: 20 times the number using the Internet in 1995

c in 2005?

Idea: 54 times the number using the Internet in 1995

- 2 What is the percentage of Internet usage that is female? Idea: 6 more than the square of 6

- 3 How many people in the United States are mobile phone owners?

Idea: 200 million – 1 million

- 4 a How many computers are there in all the government schools in Victoria?

Idea: $6731 \times 27 + 5000$

b Of these, how many are used by teachers? Idea: $6731 \times 5 - 195$

- 5 a How many tonnes of old computers are dumped each year in the United States?

Idea: $11 \times 20 \times (5 \times 2)^3 \times 1000$

b If 11% of the dumped computers are recycled, how many is that?

- 6 The cost of sending a 40-page document across the world is 12 cents. The cost of faxing it is 80 times as much and the cost of a courier is 400 times as much. How much does it cost to fax and to courier this document?

- 7 According to the US Census Bureau, the population of the world in 2006 was estimated to be 6 522 873 000.

a There are 78.3 computers for every 1000 people. How many computers are there in the world?

b One-third of the population has never made a telephone call. How many people have never made a telephone call?

- 8 There were approximately 816 million people in Africa in 2001. Convert the following proportions to numbers of people in Africa.

a 1 in 4 have a radio so people have a radio

b 1 in 13 have a TV so people have a TV



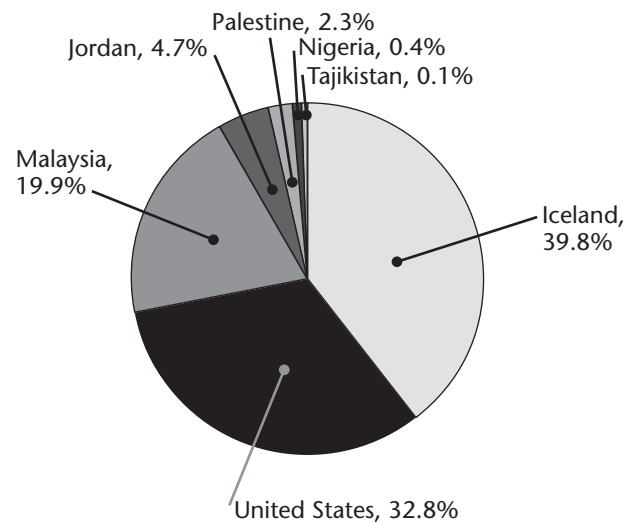
- c 1 in 35 have a mobile phone so people have a mobile phone
- d 1 in 40 have a fixed line phone so people have a fixed line phone
- e 1 in 130 have a computer so people have a computer
- f 1 in 60 uses the Internet so people use the Internet

- 9 As people age, some suffer hearing loss and need to wear hearing aids. Some hearing aids don't work very well near mobile phones. Of the hearing aids tested by a laboratory, how many could not be used near mobile phones?

Idea: $18 \times \frac{\frac{3}{4} \times \frac{5}{8}}{\frac{3}{4} \times \frac{9}{16}}$

- 10 The percentage of Internet users varies from country to country and some are shown in the graph.

- a Which country has the most users?
- b Which country has the fewest users?
- c What percentage of users is there in the United States?
- d What percentage of users are in Nigeria and Tajikistan combined?



- 11 Modern machinery can be rather noisy, which can cause hearing loss in those using it. This table shows some typical noise levels.

The hearing damage depends on the noise level (measured in decibels or dB) and the length of time the person is exposed to the noise.

- a It is said that a sound level of 140 dB would deafen the average person in a matter of months. From the table, which activities would be that dangerous?
- b Medical studies indicate that the risk associated with 85 dB over a working lifetime will produce an unacceptable hearing loss. From the table, which activities would be that dangerous?
- c In France, portable MP3 music players are limited by law to the 100 decibels level. Is that a sensible limit given the information in the table?

Activity	Noise level (decibels)
Person breathing	10
Pneumatic drill	95
Jet plane	140
Grinding machine	85
Paper-making machine	95
Road grader	94
Busy highway	70
Disco	100
Drummer in a band	140
USA factory limit	90
New Zealand factory limit	45



Our system of numbers is decimal and uses the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9. Computers use the binary system of numbers and so use the digits 0 and 1. This is because the computer is really a series of switches that allows the electricity to be on or off, or to go one way or another way.

- 1 Decimal numbers are built using the idea of place value.

	Thousands	Hundreds	Tens	Units	Tenths
a	6	4	3	2	5
b		4	6	1	0
c	5	3	1	0	1
d	6	0	0	0	3
e		2	0	0	0

Complete these number sentences in words for each row in the table.

- a 6 thousand, 4 hundred and = $6 \times 10^3 + 4 \times 10^2 + \dots$
- b 4 hundred and = $4 \times 10^2 + \dots$
- c 5 thousand = $5 \times 10^3 + \dots$
- d 6 thousand = $6 \times 10^3 + \dots$
- e 2 hundred = $2 \times 10^2 + \dots$

- 2 Find the sum of the numbers listed in the table in Question 1.

- 3 For decimal numbers, the columns expressed as powers of 10 are:

$10^3 = 1000$				
---------------	--	--	--	--

- 4 If for binary numbers the columns are powers of 2, complete these columns.

$2^3 = 8$			
-----------	--	--	--

- 5 Complete the number sentences on the next page for each binary number in the table and hence find its value.

	16	8	4	2	1
a	1	0	0	1	1
b		1	1	0	1
c			1	0	1
d				1	1





a 1 sixteen, 0 eights, 0 fours, and one
 $= 16 + 0 + 0 + + =$

b 1 eight, 1 four, two and $= 8 + 4 + + =$

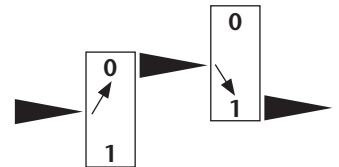
c 1 four, and $= 4 + + =$

d two and $= + =$

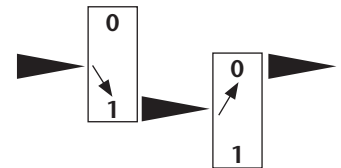
6 Complete the following table using only two digits, 0 and 1.

Decimal number	Binary number
1	1
2	10
3	11
4	100
5	101
6	110
7	111
8	1000
9	
10	
11	
12	
13	
14	
15	
16	

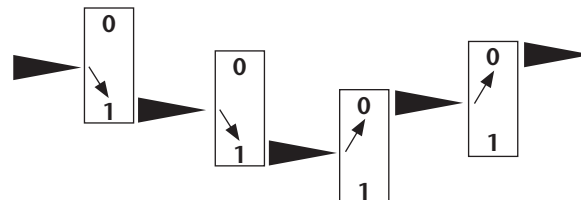
7 The reason computers generally use a binary system with only the digits 0 or 1 is that the computer circuits are made up of many switches. These switches allow the electricity to go one way (1) or the other way (0). Here is electricity being first switched to 1 and then switched to 0. This gives the number 01.



a Which binary number is given by this switch?



b Which binary number is given by this switch?



8 Here is a computer circuit taking the electricity from point X to one of the places called addresses.

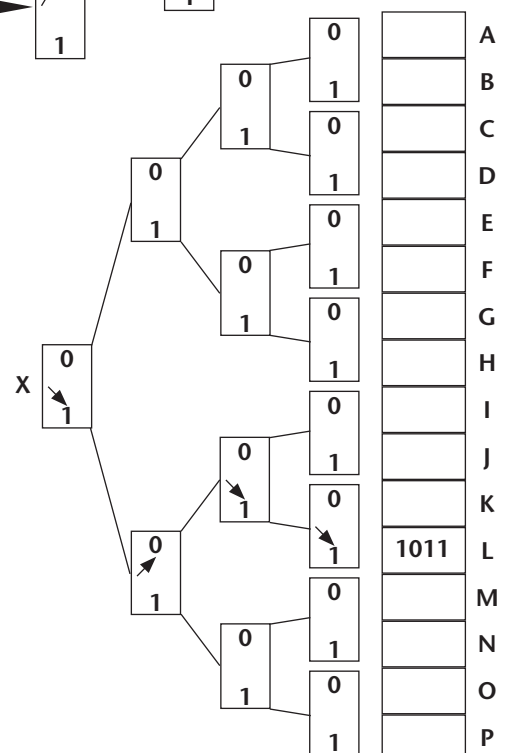
If you follow the line that the electricity would go if instructed using the binary code 1011, you will see that it ends up at the address L. Use the diagram to find the addresses from the following instructions.

a 0000

b 0010

c 1001

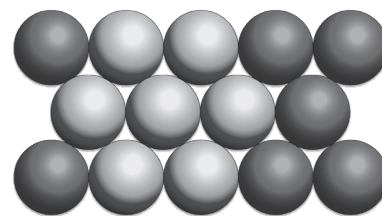
d 1111





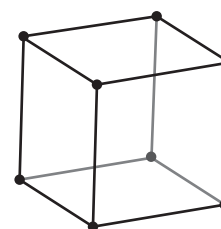
A crystal is made up of a set of atoms arranged in an organised way that repeats periodically. It could be said that the whole of civilisation stands on the crystal structure of steel and concrete. Other uses include diamonds for cutting, metal crystals for turbine blades and carbon graphite for lubrication. Of course, gems are prized for their appearance.

- 1 The spheres shown are packed together in a single layer. Seven of the spheres have been highlighted.



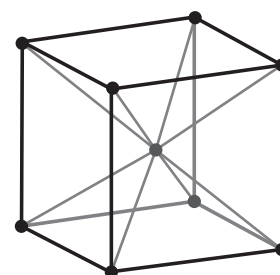
- What shape do the highlighted spheres form?
- Join the centres of the highlighted spheres and label all the angles formed.

- 2 Common salt (sodium chloride) crystals come in the form of a simple cube of atoms, shown here joined by lines. If the side length of the cube is 1 unit, find:



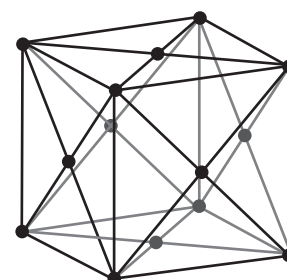
- the volume of the cube
- the number of atoms in the cube
- the length of the diagonal of one of the faces of the cube
- the length of the body diagonal (from, say, the front bottom left to the back top right) of the cube
- If the cube is repeated in every direction, to how many atoms is each atom joined?

- 3 Some metal crystals come in the form of a simple cube of atoms with an extra atom in its very centre, shown here joined by lines. This structure is called body-centred cubic. If the side length of the cube is 1 unit, find:



- the number of atoms in the cube
- the length of the body diagonal (from, say, the front bottom left to the back top right) of the cube
- the distance of the central atom from each of the corner atoms
- how many atoms each central atom is joined to.

- 4 Some crystals such as gold come in the form of a simple cube of atoms with an extra atom at the centre of each face, shown here joined by lines. This structure is called face-centred cubic. If the side length of the cube is 1 unit, find:

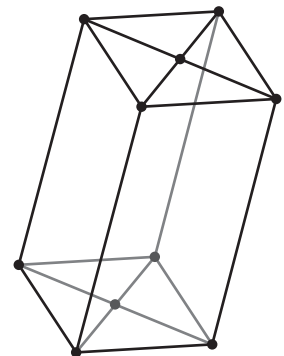


- the number of atoms in the cube
- the length of the diagonal of one of the faces of the cube
- the distance of each central atom on the face from each of the corner atoms.



- 5** Some crystals such as sulfur come in the form of a rectangular prism of atoms with an extra atom at the centre of the top and bottom faces, shown here joined by lines. This structure is called base-centred orthorhombic. If the side length of the base is 1 unit and the height is 2 units, find:

- the number of atoms in the rectangular prism
- the length of the diagonal of one of the faces of the square base
- the distance of each central atom on the face from each of the corner atoms
- the length of the body diagonal (from, say, the front bottom left to the back top right) of the prism
- the volume of the rectangular prism.

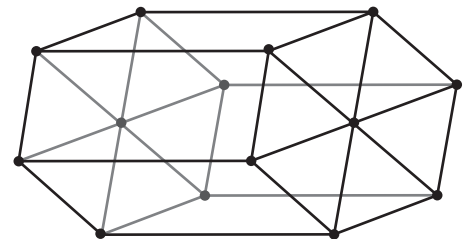


- 6** Some crystals such as graphite (carbon) come in the form of a hexagonal prism of atoms with an extra atom at the centre of the top and bottom faces, shown here joined by lines. This structure is called hexagonal. If the side length of the hexagon is 1 unit and the height is 2 units, find:

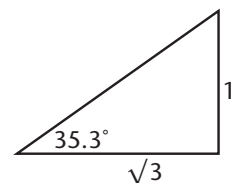
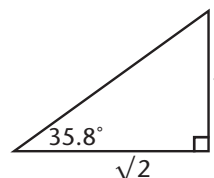
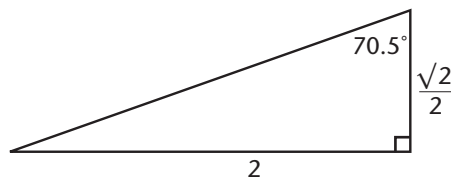
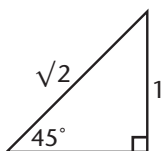
- the number of atoms in the hexagonal prism
- the distance of each central atom on the hexagonal face from each of the corner atoms

If each triangle in the base of side length 1 unit has an area of approximately 0.433 unit^2 , find:

- the approximate area of the hexagonal base
- the approximate volume of the hexagonal prism.



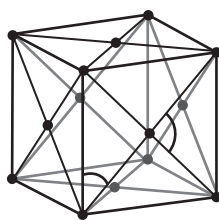
- 7** Use the triangles showing angle sizes to find the angles marked $\curvearrowright$ in each of the crystal structures shown.



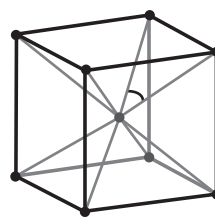
a



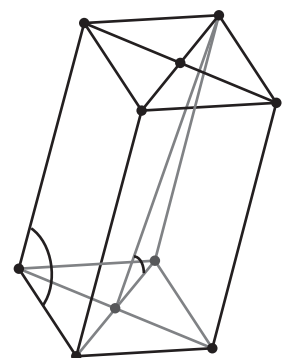
b



c

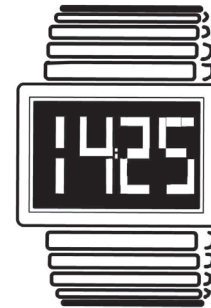
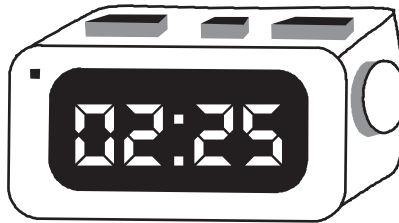
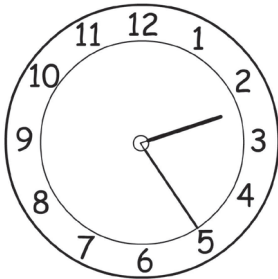


d





Clocks are used to measure time. Clocks can have a display that is analogue with time indicated by the angles between the hands on a dial; or digital as numbers on a 12-hour or 24-hour cycle. A portable clock is called a watch, while a highly accurate clock used for technical measurements is called a chronometer.



- 1 Through how many degrees does the second hand of an analogue watch travel when the time changes by:

- a 1 minute?
- b 3 minutes?
- c $2\frac{1}{4}$ minutes?

- 2 What will my digital watch read if the time is:

- a a quarter to nine in the morning?
- b a quarter to nine in the evening?



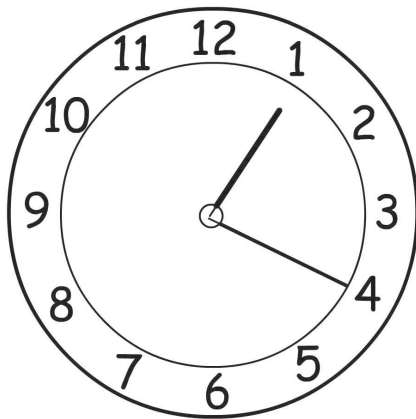
- 3 The Babylonians first used 60 as a base for units, so we have 60 seconds in a minute and 60 minutes in an hour. When I asked the other passengers how long I would have to wait for my train:
- Alice said 2 minutes less than 45 minutes
 - Bruce said 4 minutes less than 54 minutes
 - Cathy said 5 minutes less than 1 hour
 - Derek said 8 minutes more than 45 minutes.

I had to wait the longest of these times. Who was correct and how long did I have to wait?

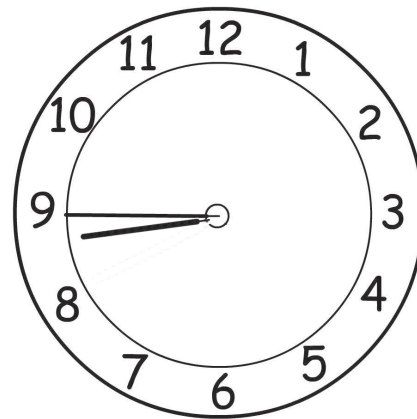
- 4 My three-year-old brother was playing with my analogue watch and turned the hands backwards and forwards. At the beginning it was reading 3 o'clock. What time was it when I stopped him if he had turned the minute hand forward 20 minutes, then back 40 minutes, then back 20 minutes, then forward 50 minutes and then back 10 minutes?
- 5 This time my three-year-old brother was playing with my digital watch and turned the numbers backwards and forwards. At the beginning it was reading 16:04. What time was it when I stopped him if he had turned the numbers forward 27 minutes, then back 33 minutes, then back 15 minutes, then forward 59 minutes and then back 3 minutes?



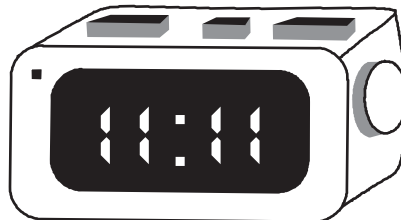
- 6 a What is the angle between the minute hand when the time is 20 past 1 and the minute hand when the time is half past 1?



- b What is the angle between the minute hand when the time is 8:45 and the minute hand when the time is twenty past 5?



- 7 What is the angle between the *hour* hand when the time is a quarter past 4 and the *hour* hand when the time is a quarter past 5?
- 8 The digital clock in the kitchen shows am and pm with the time indicated by two digits for the hour (from 00 to 12) and two digits for the minutes (from 00 to 59).



- a At what times are all the digits the same? (See the picture above for an example.)
- b i What digits can appear in the first place on the clock? How many different numbers is that?
- ii What digits can appear in the second place on the clock? How many different numbers is that?
- iii What digits can appear in the third place on the clock? How many different numbers is that?
- iv What digits can appear in the fourth place on the clock? How many different numbers is that?
- v If you multiply the number of different numbers from the four parts above, you will find how many different times the clock can indicate.
- c At how many times will all the digits *not* be the same?



Numbers in digital displays are usually made using rows of lights, so the numbers have no curves. As a result, some look like letters when they are turned upside down.

- 1 Here are some numbers on a calculator screen.

0 1 2 3 4 5 6 7 8 9

- a Turn this page upside down. Which of the numbers look like a letter of the alphabet?
For example, when you turn

4

upside down it becomes

h

and looks like an 'aitch'.

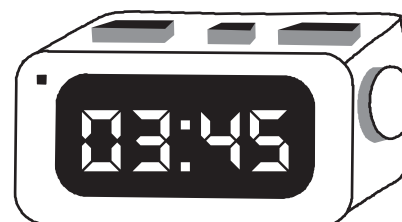
- b Use Part a to fill in the following table.

Number	0	1	2	3	4	5	6	7	8	9
Letter (if any)										

- c What digits would you enter so that the calculator spelt SHELL when you turn it upside down?
- d Write down as many words as you can using the letters found in Part b.
- e Use the calculator to find the sum of 2, 5 and 7. What does it spell when you turn the calculator upside down?
- f *Without* using a calculator, multiply 1 772 569 by 3. What word would be formed if you turned a calculator that showed this number upside down?
- g For each of the words you found in Part d, make up one arithmetic problem that will give the word when you turn the calculator upside down.

- 2 Digital clocks, both 12-hour and 24-hour, can reveal words when you turn them upside down.

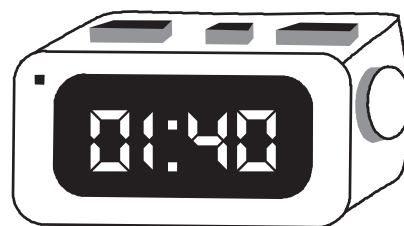
- a A digital clock reads 3:45 pm. What is spelt when the clock is turned upside down?
- b What time will it be:
- 8 minutes earlier?
 - 41 minutes earlier?
- c What will the clock spell upside down in each of the cases in Part b?





3 a A 24-hour clock reads 0140 hours.
What is spelt when the clock is turned upside down?

- b** What time will it be:
- i 3 minutes earlier?
 - ii 2 hours and 30 minutes later?
 - iii 11 hours and 11 minutes later?



c What will the clock spell upside down in each of these cases?

4 a Could a 24-hour digital clock spell out the word SHOE when turned upside down? Give reasons for your answer.

b Could a digital clock spell out SILO when turned upside down? Give reasons for your answer.

5 A 24-hour digital clock shows the time from 00:00 to 23:59.

a What digits can appear in the last position?

b What digits can appear in the first position?

c When the clock is turned upside down, what letters can the words start with?

d When the clock is turned upside down, what letters can the words end with?

e Write down as many words as you can using the allowed digits of a 24-hour clock.

6 In the following grid, find the listed words and cross them out in the grid. Words may be horizontal, vertical or diagonal. Which letter is left over?

M	E	L	O	D	I	C	G
A	A	I	R	O	T	O	R
T	R	C	H	A	R	M	A
H	S	H	H	X	A	P	P
S	E	E	P	I	I	U	H
O	V	N	O	O	N	T	E
C	E	P	R	O	V	E	N
K	N	I	T	T	E	R	S

Word list

Machines	Rotor	Graph
Maths	Train	Melodic
Seven	Seep	Pi
Proven	Computer	Sock
Knitters	Hens	Noon
Charm	Lichen	Ears



Numbers can be expressed correct to a particular number of decimal places or a particular number of significant figures. The number of digits we use to describe a measurement is the number of significant figures. Numbers with too many decimal places or too many significant figures are rounded off.

1 Count the number of significant figures in each of the following:

- a 23
- b 203
- c 230
- d 23.00
- e 2003.0

Rules for counting significant figures

- Count all zeros within a number as significant (2305 and 2.005 have 4 sig figs).
- Count all zeros to the right of the decimal point as significant (56.00 and 5.050 have 4 sig figs).
- Don't count zeros that keep the decimal point in place as significant (3200 has only 2 sig figs).

2 Write the following correct to 3 significant figures:

- a 4852
- b 72.76

3 Write 5.048 correct to 2 significant figures.

4 Write the number 2.477 correct to:

- a 2 decimal places
- b 2 significant figures

5 Write the number 0.0275 correct to:

- a 2 decimal places
- b 2 significant figures

6 a Write correct to 2 significant figures:

- i 2.52
- ii 109

b Hence find an approximate value of 2.52×109 .

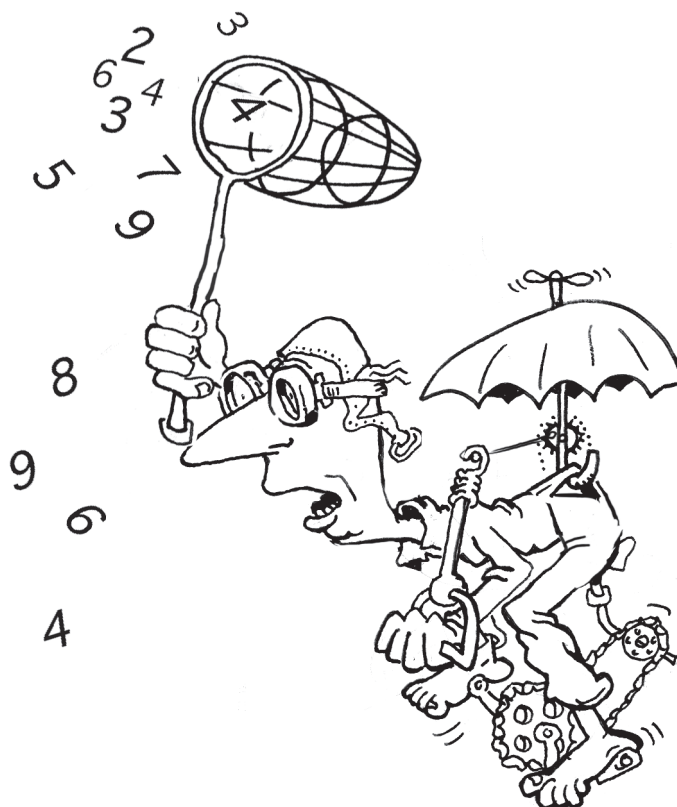
7 a Write correct to 2 significant figures:

- i 0.3267
- ii 0.0599

b Hence find an approximate value of $0.3267 \div 0.0599$.

Rules for Rounding:

Numbers with too many decimal places or too many significant figures are rounded off. Remember, all digits after the rounding digit are dropped. If the digit to the right of the rounding digit is equal to or more than 5, add 1 to the rounding digit.





8 You measured a table to be 2.4324 m long with a ruler that can only measure to centimetres. To how many significant figures can you be confident? What should you have said the length of the table was?

9 You added 0.485 g of sugar to 125.0 g of liquid.

a What was the total weight of the solution?

Since you can only be confident of the number of digits in the least accurate number, the final answer can only be as accurate as that number.

b How many significant figures has:

i the sugar measurement?

ii the liquid measurement?

c Which is the least accurate?

d State the total weight of the solution to the correct accuracy.

10 Your friend in the United States told you that the temperature outside her house was measured to be 39.2°F.

Using the formula:

$$^{\circ}\text{C} = \frac{5}{9} (^{\circ}\text{F} - 32)$$

find the temperature in degrees Centigrade to the correct number of significant figures.

11 In the following grid, find the listed numbers and circle them in the grid. Numbers may be horizontal, vertical or diagonal. Which number is left over?

3	4	1	2	8	2
7	1	9	6	4	2
9	8	4	6	0	4
7	2	7	1	8	3
1	9	2	6	5	3
5	2	2	2	2	9

Number list

314 159

27 183

2 684

224

3 412

9 222

12 468

339

789

1 603

2 222

59 764

229



Google is one of a number of search engines allowing us to search the Internet. Such search engines have been criticised because there can be a lack of discrimination between data (factual information) and information (data that has been interpreted).

- 1
 - a When was the first search engine invented and what was its name?
 - b In which year were the most search engines invented?
 - c In which year was Google invented?
- 2 In 1995 the search engine Lycos had indexed 90% of the 10 million documents on the Internet. How many had Lycos indexed? (Indexing is like making a library catalogue of the documents.)
- 3 HotBot can index 10 000 000 pages of documents per day. How many pages can it index per year?
- 4 It has been estimated that in 1999 there were nearly 600 million pages on the Internet. The search engine AltaVista had only indexed 150 million pages at that time.
 - a What percentage of the whole web had AltaVista indexed?
 - b The search engines Excite and Lycos had only indexed 50 million pages. What percentage of the web was left to be indexed in 1999?
- 5 Google receives 2000 search requests every second. How many searches per day is that?
- 6 The Google index is 2 Terabytes. How many items are there in the index if Tera means 10^{12} ?
- 7
 - a How many web pages does Google have stored?
Idea: $3.1 \times 10^3 \times 10^3 \times 10^3$
 - b How long in years would it take to search all these web pages by hand if it takes one minute per page?

History of Search Engines

1990	Archie
1991	Gopher
1993	Veronica
1994	WebCrawler
1994	Yahoo!
1994	Excite
1995	Lycos
1995	AltaVista
1995	HotBot
1995	MetaCrawler
1997	Excite
1997	Northern Light
1998	Google



- 8 Google has 10 000 computers each containing two hard disks holding 40 Gigabytes of information.
 - a How many Gigabytes of information is that?
 - b How many bytes of information is that if Giga means 10^9 ?
- 9 A search on the word Mathematics yielded 504 000 000 results in 0.21 seconds. At that speed, how many results would be found in $\frac{1}{10}$ second?
- 10 Google charges a minimum of 5 cents for every advertisement clicked on. If 5% of the 2000 searches per second result in a link to an advertisement:
 - a how much does Google earn every second?
 - b how much does Google earn per year from advertising links?
- 11 a The name Google is based on the mathematical word 'googol' which has the value one followed by how many zeros?

Idea: $\frac{3}{8} \times \frac{1}{4} \times 10 \frac{2}{3} \times 100$

- b Write a googol as a power of 10.
- c A googolplex is one with a googol zeros.
Write a googolplex as a power of a power of 10.
- 12 Google is used in how many languages?

Idea: $100 - \frac{2}{3} \div \frac{2}{9}$

- 13 In 2005, a company called Double Click had a good program for managing the advertising on the Internet. It was privatised for \$US 1.1 billion dollars.

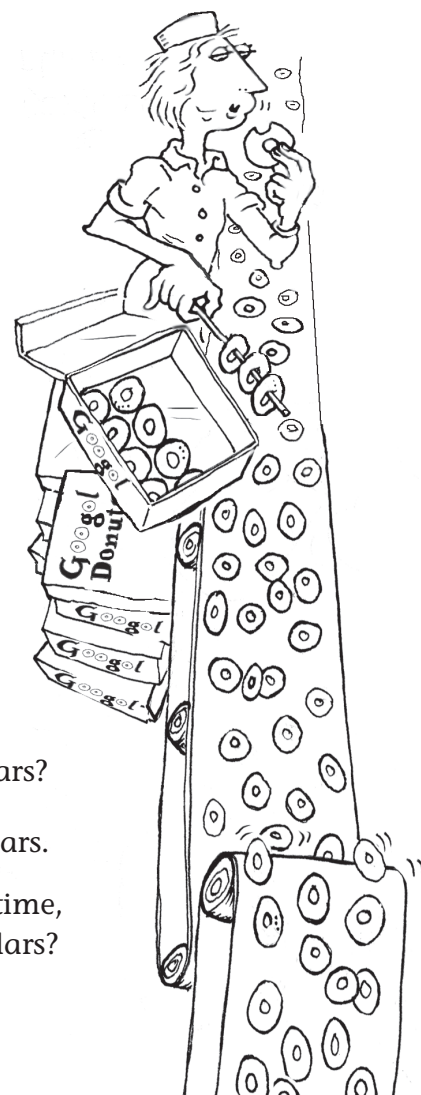
- a If a billion dollars is a thousand million dollars and at that time one Australian dollar was worth 65.83 US cents, how much was that in Australian dollars?

In 2006, Google bought the company UTube for \$US 1.76 billion dollars.

- b If one Australian dollar was worth 75.89 US cents at that time, how much did UTube cost Google in Australian dollars?

In 2007, Google bought Double Click for \$US 3.1 billion dollars.

- c If one Australian dollar was worth 82.37 US cents at that time, how much did Double Click cost Google in Australian dollars?
- d How much more was Double Click worth in 2007 than it was in 2005 (in Australian dollars)?





Estimating can be used to tell if the answer the calculator gave is sensible. You can also estimate amounts of money, length, time, distance, and so on. Rounding makes for numbers that are easy to use in an estimate.

- 1
 - a Round 215 to the first digit, the hundreds.
 - b Round 179 to the first digit, the hundreds.
 - c Use these answers to estimate the sum of 215 and 179.
 - d What is the actual sum of 215 and 179?
- 2
 - a Round 4.85 to the first digit, the units.
 - b Round 0.28 to the first digit, the tenths.
 - c Use these answers to estimate the sum of 4.85 and 0.28.
 - d What is the actual sum of 4.85 and 0.28?
- 3
 - a Round 72 to the first digit, the tens.
 - b Round 0.46 to the first digit, the tenths.
 - c Use these answers to estimate the product 72×0.46 .
 - d What is the actual product 72×0.46 ?
- 4
 - a Find an estimate of 6.3×10.2 using first digit rounding of each number.
 - b What is the actual product 6.3×10.2 ?
- 5
 - a Find an estimate of $56.7 \div 1.8$ using first digit rounding of each number.
 - b What is the actual quotient $56.7 \div 1.8$?
- 6
 - a Find the approximate value of $33.3 \times 9.30 \div 22.2$ using first digit rounding of each number.
 - b What is the actual value of $33.3 \times 9.30 \div 22.2$?
- 7 Find $\frac{1}{2}$ of $56 + 12 \times (32 - 17) \div 5 - 12$
- 8 The calculated value of $4.35 \times 0.38 + 7.6 \times 0.58$ was 64. Use first digit rounding of each number to show that the answer is wrong by a factor of 10.



Rounding rules

For whole numbers, all digits to the right of the rounding digit become zero.

For decimal numbers, all digits after the rounding digit are dropped.

In each case, if the digit to the right of the rounding digit is equal to or more than 5, add 1 to the rounding digit.

Order of operations check!

Brackets are first,

Of is second,

Division and Multiplication,

then Addition and Subtraction working left to right



- 9** Use first digit rounding to find if the value of $6.26 \times 2.3 + 92 \div 21 \times 0.049$ is more or less than 10.
- 10** Use first digit rounding to find whether $5.97 \times 4371 - 25\,253$ should give a positive or negative answer. Use a calculator to check your answer and explain the result.
- 11** When finding the radius of a circle of circumference 256 cm I needed to divide by 2π where $\pi \approx 3.1416$. My calculator gave the result 4.074 cm. Use first digit rounding to check if this is likely to be correct.

- 12** A single optic fibre can carry a large number of simultaneous telephone calls.

a Work out approximately how many that is by pressing the following keys in order.



b What answer do you get if you leave out the equals sign after pressing the 9? Is this the correct answer?

- 13** Find the exact values of the following without a calculator:

a $39\,554\,639\,751\,112 \times 7$

b $\frac{3}{7} \times \frac{1}{12} \div 6\frac{1}{4}$

- 14** Using a calculator may be tricky. Here are some of the trickier ones for you to try:

a $8 - 2 \times 3$

b $(8 - 2) \times 3$

c -2^2

d $\frac{4 \times 2 - 3}{\sqrt{4+4}}$

e $5^2 5^3$

f $\sqrt{4} + \sqrt{9}$

- 15 a** Use your calculator to perform the following multiplications:

i $321 \times 7 \times 11 \times 13$

ii $842 \times 7 \times 11 \times 13$

b What would you get if you multiplied $539 \times 7 \times 11 \times 13$?

- 16 a** Use your calculator to perform the following multiplications:

i $32 \times 3 \times 7 \times 13 \times 37$

ii $57 \times 3 \times 7 \times 13 \times 37$

b What would you get if you multiplied $98 \times 3 \times 7 \times 13 \times 37$?

- 17 a** Use your calculator to perform the following multiplication: $5869 \times 11 \times 9091$

b What would you get if you multiplied $3758 \times 11 \times 9091$?

c Ask a friend to think of a 4-digit number and multiply it 11, then 9091, and give you the answer. Amaze them by correctly guessing the number.

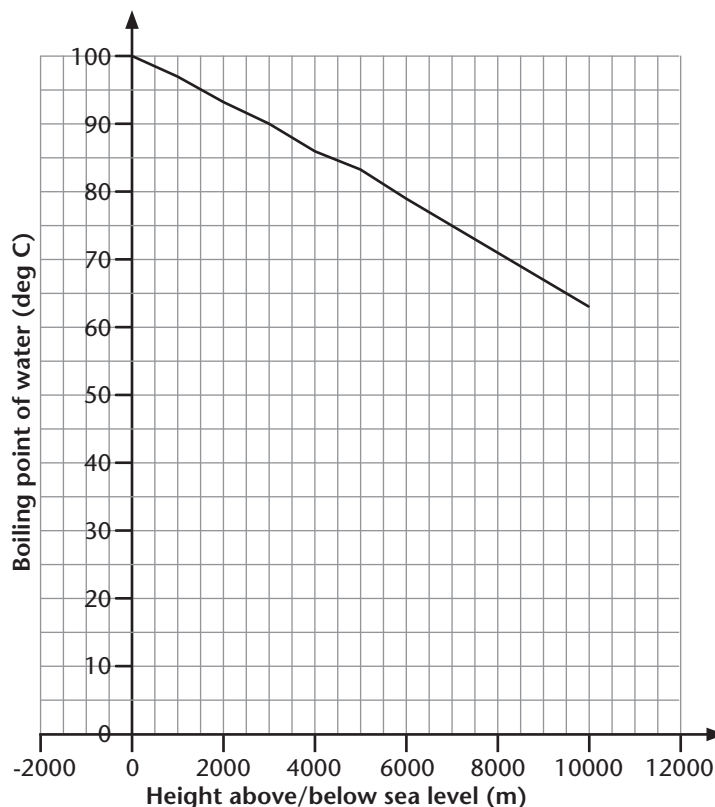


Because air pressure is different at different heights above sea level, at high altitudes it can be difficult to breathe, water boils at different temperatures and cooking becomes more difficult.

- 1 What is the boiling point of water at sea level? Idea: $5^2 \times 4$
- 2 What is the boiling point of water on top of Mount Everest (8850 m above sea level)? Idea: $25 \times 7 \times 2 \div 5$
- 3 What is the boiling point of water in Lake Eyre (12 m below sea level)? Idea: $17.02 + 24.39 + 57.6 + 1.05$



- 4 Water boils at different temperatures, depending on how far above or below sea level you are. This is shown in the graph.



- a Does the boiling point of water go up or down as you go up a mountain?
 - b Does the boiling point of water go up or down as you go below sea level?
 - c From the graph, find how hot the water is when it boils if it is:
 - i at sea level.
 - ii 3000 m above sea level.
 - d Although airliners fly at about 6710 m, the cabin pressure is usually set for 1830 m. At what temperature would water boil inside an airliner?
 - e There is still enough oxygen to breathe when you are at an altitude of 4575 m. If you were camping at that altitude, at what temperature would your water boil?
 - f Place a ruler over the graph. Is it a straight line? Estimate the boiling point of water on the top of Mt Kosciuszko (2228 m).
- 5 Because water boils at a lower temperature at higher altitudes, food takes longer to cook. My friend Amanda lives in Colorado at about 1600 m above sea level where the boiling point of water is 95°C.



- a Usually she likes to make her tea with boiling water so that it is strong. How will the altitude affect her tea?
- b For a hard-boiled egg the egg must be boiled in water at 100°C for 5 minutes. It has been found that the cooking time can be calculated using the formula:

$$\text{Cooking time at a particular altitude} = \frac{100 \times \text{Usual cooking time}}{\text{Temperature for boiling at that altitude}}$$

So for a hard-boiled egg to be cooked in Colorado the cooking time will be $\frac{10 \times 5}{95} = 5.3$.

Amanda liked her egg boiled for 3½ minutes when she lived at sea level. How long should she cook it in Colorado?

At higher altitudes the air pressure is lower and this affects the way you have to cook cakes and biscuits.

Amanda has found that at her altitude:

- heat rises more from the bottom of the oven because there is not enough air pressure to stop it
- cakes rise too quickly and too high, causing the cakes to fall, so she must use 5% more flour and reduce the baking powder by half
- cakes rise too high and the mixture overflows the tin so she must fill pans half full
- cakes can come out undercooked because of the lower temperature, so she must bake at 15°C higher but 20% more quickly
- liquids evaporate more quickly and the cake dries out, so she must use 20% more water or add 2 tablespoons of water to the recipe
- sugar becomes more concentrated, so for each cup of sugar in the recipe, she must decrease it by 3 tablespoons (a tablespoon of white sugar weighs 20 g and a tablespoon of brown sugar weighs 15 g).

- c Complete the table to show what Amanda must do with her cookie recipe.

Chocolate-chip cookie recipe

- 250 g plain flour
- 1 teaspoon baking powder
- 1 teaspoon salt
- 125 g butter, softened
- 110 g white sugar
- 85 g brown sugar
- ½ teaspoon vanilla extract
- 1 egg
- 125 g chocolate chips
- 125 g chopped nuts

1. Mix the flour, baking powder and salt in small bowl.
2. Beat the butter, white sugar, brown sugar and vanilla in large bowl.
3. Add egg, beat well then gradually beat in flour mixture.
4. Stir in chocolate and nuts.
5. Drop tablespoonsful onto an ungreased baking tray.
6. Bake in 200°C oven for 10 minutes or until golden brown.
7. Let stand for 2 minutes; then remove to wire racks to cool completely.

Recipe	Change
250 g plain flour	
1 teaspoon baking powder	
110 g white sugar	
85 g brown sugar	
0 water	
Bake at 200°C	
Cooking time 10 min	



Mobile telephones have now become part of everyday life for many people. This means individuals can be contacted 24 hours per day.

- 1 a How many mobile phones were in use in Australia in the year 2000?

Idea: $9 \text{ million} - 107 \times 19 \times 9 \times 4 \times 6$

- b If 98.7% of mobile phones were digital, how many were digital?

- 2 According to the Australian Bureau of Statistics, in the year 2000, 60% of drivers in Western Australia had mobile phones and there were approximately 1 429 723 people of legal driving age who actually drive. Approximately how many mobile phones were owned by drivers?

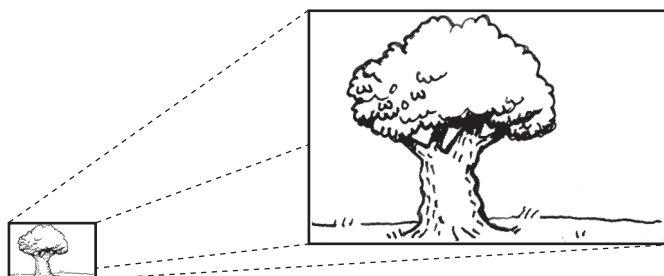
- 3 It has been suggested that 85% of mobile phone owners make calls while driving and 27% use their phones at some time during a trip. If there are 5 136 523 drivers in Australia, how many could be expected to:

- a make calls while driving?

- b use their phones at some time during a trip?

- 4 The risk of a learner driver being involved in a fatal car crash is 10 times as much as for an experienced driver. This risk is increased by a factor of 4 when the driver is using a mobile phone while driving. By what factor is the risk of a learner driver being involved in a fatal car crash increased if they are using a mobile phone?

- 5 'Optical zoom' is like magnifying a picture. So if a picture is magnified six times it is usually described as having $6\times$ zoom; the size of the image is six times larger when zoomed than when it was not zoomed. The word 'optical' just means that it uses the camera lens, not a digital system. The camera in Elly's mobile phone has $12\times$ optical zoom but her sister Bea's has only $8\times$ optical zoom.



- a What would be the dimensions of the zoomed image of a tree if the unzoomed dimensions were 1.5 cm by 0.5 cm:
- on Elly's camera?
 - on Bea's camera?

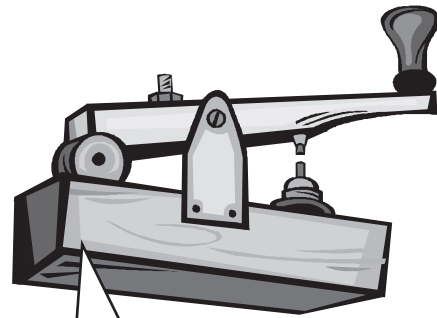


- b** What would be the *areas* of the zoomed images of the tree in each case:
- on Elly's camera?
 - on Bea's camera?
- c** The ratios of the dimensions of the zoomed image to the unzoomed image are 12 : 1 and 8 : 1 respectively. What are the ratios of the areas of the zoomed image to the unzoomed image?

- 6** A driver's reaction time is the time taken to apply the brakes (or swerve) after seeing danger. For average drivers, the reaction time is about 1.5 seconds.
- If the driver is travelling at 60 km/h at the instant he applies the brakes, how far in metres will he travel during that 1.5 seconds?
 - The reaction time is said to be increased by about 0.5 seconds when the driver is using a mobile phone while driving. What is the reaction time of a driver using a mobile phone?
 - For a driver travelling at 60 km/h, how much further will he travel during the extra time?

- 7** When you receive an SMS (Short Message Service) message on your mobile phone, you hear a sound like 'dit dit dit da da dit dit dit'. This is Morse code for the letters SMS. Morse code is made up of a series of dots and dashes and was used to send messages over telegraph lines during the last century. Skilled coders could send messages at 30 words per minute and could even tell who was sending the message by the way the dots and dashes sounded!

- ... - - - ... is a famous distress signal in Morse code. What does it say?
- What is Morse code for the word 'Maths'?
- Morse chose the letters that occur most frequently in English to have the shortest codes. Which two letters are the most commonly used?
- If a dot lasts $\frac{1}{8}$ second, use the rules to calculate how long it takes to send the word 'Maths'.



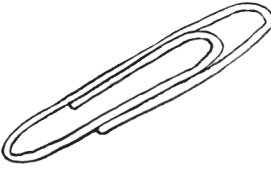
A .-	B -..	C -..	D -.	E .
F ..-	G --.	H	I ..	J .---
K -.-	L .-. .	M --	N -. .	O ---
P .-.	Q --.-	R .-. .	S ...	T -
U ..-	V ...-	W .--	X -.-.	Y -.-.
Z ---.				
1 .----	2 ..----	3 ...--	4-	5
6 -....	7 ---..	8 ----.	9 -----	0 -----
Full stop	comma .- .- .-			

Rules for sending Morse code

- A dash must last as long as three dots.
- A gap as long as one dot must be left between repeated letters.
- A gap as long as one dash must be left between different letters.
- A gap as long as five dots must be left between different words.

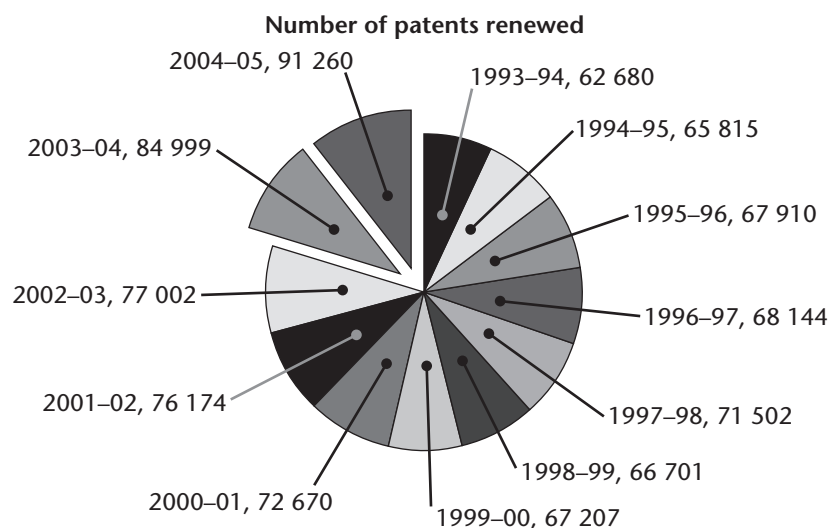


When someone invents something new they often protect the idea by 'taking out a patent'. The patent describes the invention but stops other people from making or selling it without paying for the right.

- 1 Of the 24 566 patents applied for in Australia, only 12 388 were granted.
 - a What percentage of patents was granted?
 - b If each patent application cost approximately \$7000, how much money did the patent office receive?
- 2 Disposable pairs of bamboo chopsticks joined together were invented in Japan in the late 19th century. Usually they are approximately 25 cm long. If 24 000 000 000 pairs are sold each year, how many kilometres of bamboo are needed?
- 3 Paper clips like the one pictured were invented in 1892. This paper clip is made of wire 0.08 cm in diameter and 9.85 cm long. An office uses 7550 paper clips each year. What length of wire in metres was used to make these paper clips?
 
- 4 Each year \$74 000 000 000 is paid by companies to inventors through their patents. 96% of these so-called 'royalties' are paid by companies in the developed world.
 - a What percentage of royalties is paid by companies to the rest of the world?
 - b How much is paid in royalties in the rest of the world?

(Be careful! The calculator may not be able to display enough digits.)

- 5 Patents can only be taken out for a certain period of time, then they run out and need to be renewed. The following pie chart graph shows the number of patent renewals in Australia in recent years.

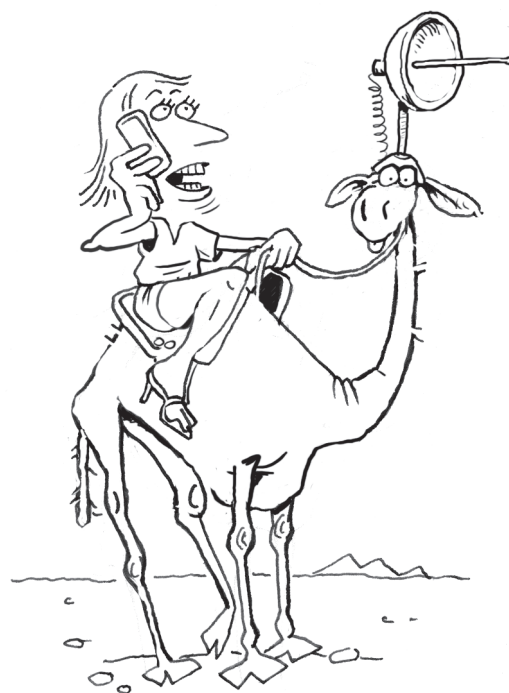


- a In which years were the most patents renewed? How many were renewed?
- b In which years were the fewest patents renewed? How many were renewed?
- c By how much did the renewals fall between 1997-98 and 1998-99?



- d** Express the fall between 1997–98 and 1998–99 as a percentage of the number renewed in 1997–98.
- e** What was the percentage rise in the years 2003–05 and was it greater or less than the percentage fall in 1997–99 found in Part d?
- 6** My analysis by country of the inventions listed on a website gave me the following results.

Country	Number of inventions
United Kingdom	35
United States	56
Sweden	21
Germany	19
Hungary	23
Canada	5
Italy	4
France	4
Croatia	4
Russia	4
Yugoslavia	6
Australia	4

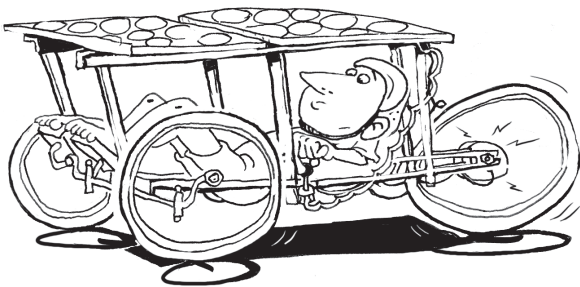


Other countries all had fewer than three each.

- a** How many inventions were counted in the listed countries?
- b** Which country had the most inventions listed?
- c** If there were 28 inventions listed for England, how many were listed for Scotland, Ireland and Wales? (The United Kingdom comprises England, Scotland, Ireland and Wales.)
- d** Italy, France, Germany, Sweden and Hungary are all members of the European Union (EU). How many inventions came from the EU?
- e** Australia has approximately 20 500 000 people, the EU has approximately 460 000 000, the United Kingdom has approximately 59 800 000 and the United States has approximately 298 000 000. Find the number of inventions per head of population in:
- i** Australia
 - ii** European Union
 - iii** United Kingdom
 - iv** United States
- f** Which of these countries has the greatest number of inventions per head of population?



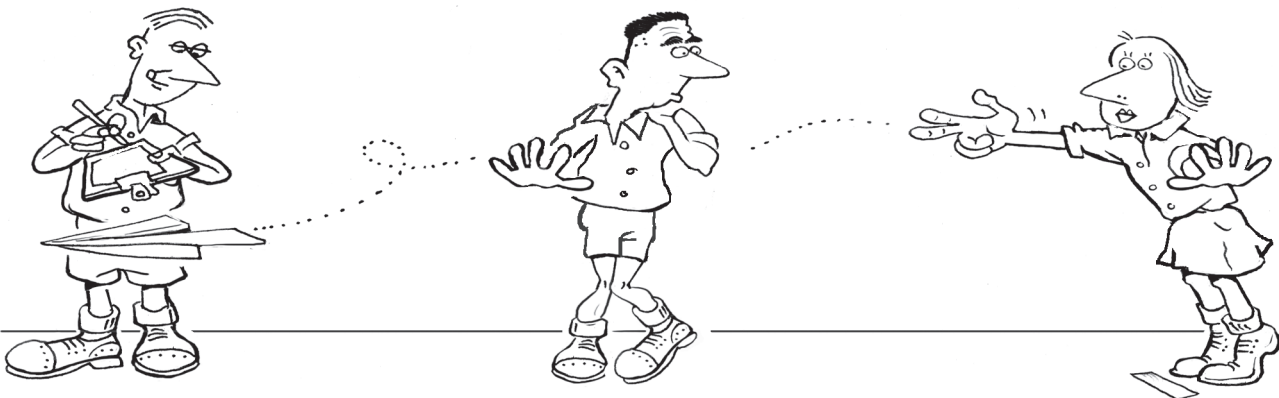
Schools make use of technology such as computers, and sometimes even require students to make their own machines such as solar cars and paper aeroplanes.



1 A Science class decided to make paper aeroplanes and test them to see how far they could fly. Their results are shown in the following table.

Name	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Distance (m)	6.4	1.8	0.03	4.3	0.3	4	1.3	3.6	2	6.3	5	5.55	1.69	1.54	0.82

- a Whose plane flew the furthest distance?
- b Whose plane flew the shortest distance?
- c The average distance flown by the 15 planes tested can be found by adding all the distances and dividing the result by 15. What was the average distance flown?
- d The median distance flown can be found by placing the distances in order from smallest to largest and taking the middle value. What was the median distance flown?



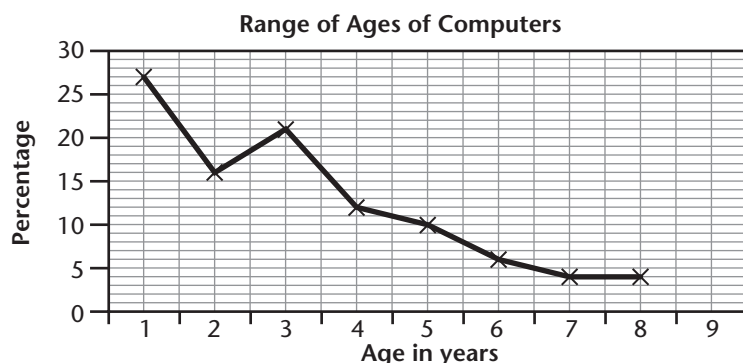
2 The teacher has a rectangular piece of cardboard and tells her class that its length is one more than three times its width, and that it has an area of 80 cm². She wishes her students to find the dimensions of the cardboard. The class makes a table of values on a computer.

Width, w (cm)	1	2	3	4	5	6	7	8
Length, l (cm)	$3 \times 1 + 1 = 4$	$3 \times 2 + 1 = 7$						
Area, l $\times$ w (cm ²)	$1 \times 4 = 4$	$2 \times 7 = 14$						

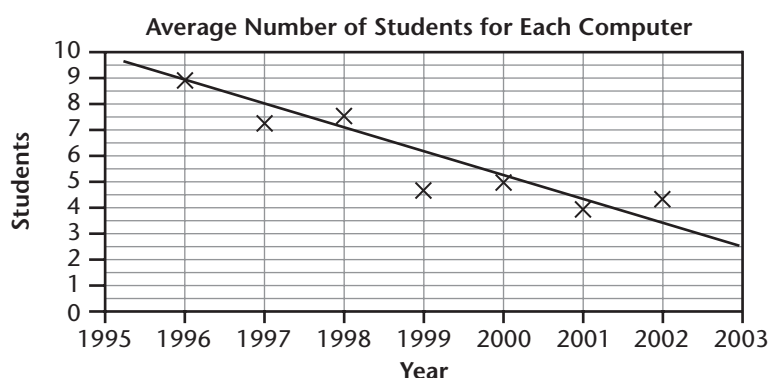
Complete the table and hence find the dimensions of the teacher’s cardboard.



- 3** The government provides computers to schools each year. Some computers are as old as 8 years, some as young as 1 year.



- a** What percentage of computers is 1 year old?
- b** What percentage of computers is 8 years old?
- c** When were there large numbers of computers bought?
- 4** The average number of students per computer in schools has been getting less as shown in the graph.
- a** Which is better, a large number of students or a small number of students?



A 'line of best-fit' has been drawn on the points. This can be used to predict past or future values by extending the line.

- b** Use a ruler to estimate the number of students per computer in 2003.
- c** Use a ruler to estimate the number of students per computer in 1995.

If y is the number of students per computer and x is the number of years past 1995, the straight-line graph can be described by the relationship $y = mx + C$ where the gradient is m and the Y -intercept is C .

- d** What is the Y -intercept?
- e** Is the gradient positive or negative? Find the gradient.
- f** Write down the equation of the line.
- g** Use the equation of the line to estimate the number of students per computer in 2004, 2005 and 2006 (that is, for values of $x = 9, 10$ and 11). Are these reasonable values? Why?

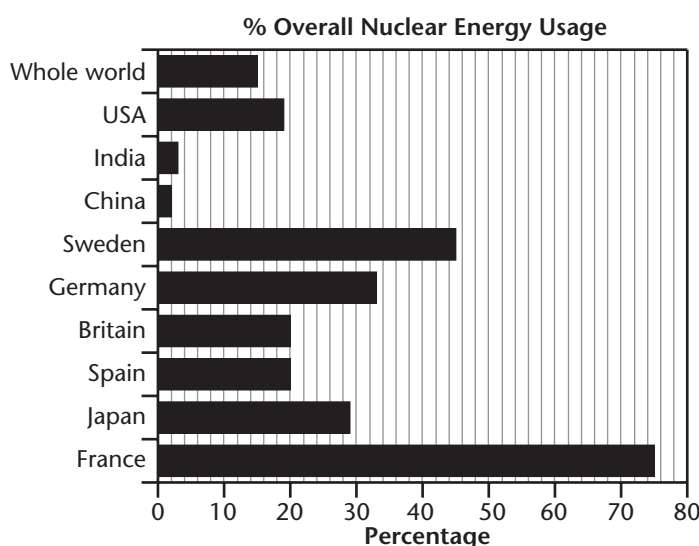
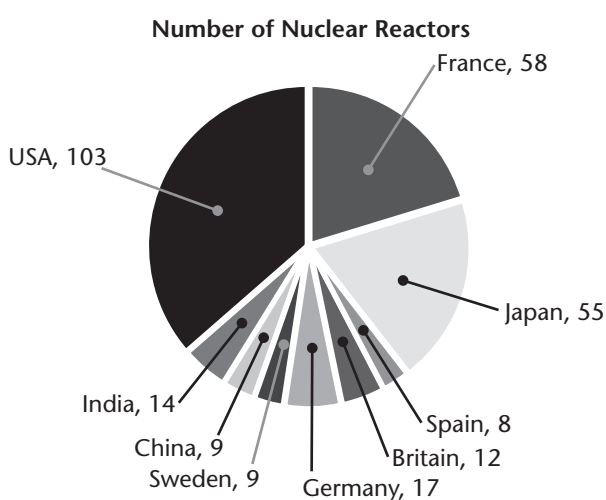


People use energy for such things as running cars and making electricity. Nuclear power is one source of energy but there are others such as coal, oil and 'green power'.

1 Nuclear reactors provide 15% of the world's electricity. In France, where there are 58 reactors, about 750 kilograms of high level waste are produced from one year's operation of a typical nuclear reactor. How much waste does France need to store each year?



2 Many countries use nuclear reactors to generate electric power. Use the graphs to answer the following questions.



- Which country has the most nuclear reactors?
- Which country has the greatest percentage of its energy coming from nuclear reactors?
- The European countries using nuclear power are France, Spain, Britain, Germany and Sweden. How many nuclear reactors are there in Europe?
- What percentage of Japan's energy is nuclear?
- Worldwide, what percentage of energy for electric power comes from nuclear reactors?
- If coal accounts for 40% of energy generation worldwide, what percentage of energy is generated by means other than nuclear or coal?

Source data for graphs: Uranium Information Centre – *The Age* 19/6/06 article by Rod Myer.

High Level Waste

In France, reactor cores are put in concrete for 50 years. The high level waste, which includes Strontium and Iodine, are stored for 40 years. After this time, the radioactivity is reduced to 0.1% of its original level.



- 3** **a** Use the definition of energy given to find the energy used when a 2000 watt heater was left on for 3 hours.
- b** The cost of producing nuclear power has been estimated to be \$60 per Megawatt hour (MWh). Power production costs for coal are approximately \$41 per MWh, gas \$47 per MWh, wind \$75 per MWh and solar \$50 per MWh.

Fabrico's factory used 50 MWh of electricity last year. Find the production cost of electricity using each of the fuels listed.

- 4** In 2001, Australia produced 2200 petajoules of energy, where a petajoule is 30 000 Megawatt hours.

- a** How many Megawatt hours of energy were produced in Australia in 2001?

In Australia, electricity is produced mainly from coal (about 75%), natural gas (about 14%) and oil (about 1%). Renewable resources such as hydroelectric, wind and biofuels (e.g. ethanol made from sugar cane) make up about 8%.

- b** How much energy in MWh was produced by:
- i coal
 - ii natural gas
 - iii oil
 - iv renewables?

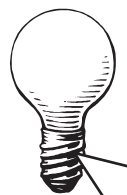
- c** If Australians used 5037 petajoules of energy, how much energy (in petajoules) needed to be imported?

- 5** Greenco sells green energy to its customers. It gets 58% of its energy from biomass, 36% from wind, 5% from hydro (water) and 1% from solar. Greenco analysed its sales in order to advertise better to its customers. Unfortunately some of the data was lost.

- a** Complete the table.

	Homes	Businesses	Total
Amount of energy sold (MWh)	183 000	298 000	481 000
Number of customers	125 000	6 000	
Amount of energy from biomass (MWh)		172 840	278 980
Amount of energy from wind (MWh)	65 880	107 280	
Amount of energy from hydro (MWh)		14 900	24 050
Amount of energy from solar (MWh)	1 830		

- b** If the advertising states that 'over 115 000 homes use electricity from biomass or hydro', would that be a correct statement?



About power

A watt hour is one watt of power used for one hour. So a 60 watt light globe running for 1 hour would use 60 watt hours of energy. If it ran for half an hour it would use 30 watt hours of energy.

Energy (watt hours) = power (watts) x time (hours)

A Megawatt hour (MWh) is a million (Mega) watts of power used for one hour.

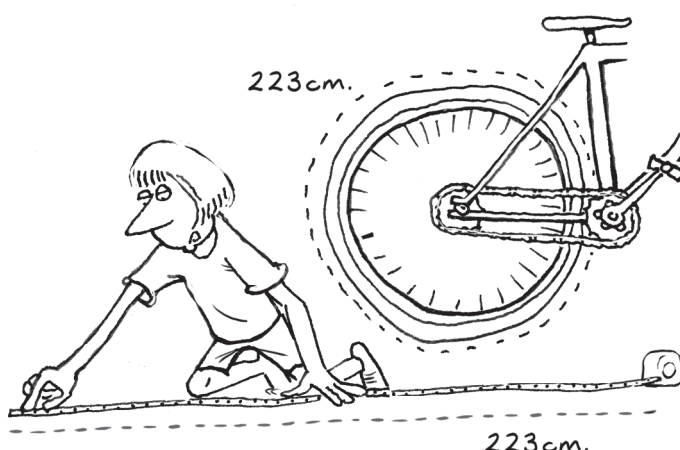
A kilowatt hour (kWh) is a thousand (kilo) watts of power used for one hour.

A petajoule (pJ) is 30 million kilowatt hours or 30 thousand Megawatt hours.



Bicycles are a means of transport as well as being ridden as a sport. Tricycles are three-wheeled children's toys.

- 1 In the 1970s, Americans bought 11 million cars each year. They bought 22% more bicycles than cars. How many bicycles did they buy?
- 2 The cost of a bike in the 1880s was approximately \$125 at a time when the average salary was \$450 per year. The average salary in Australia is currently \$54 000. How much would a bike cost now if it were the same proportion of average salary?
- 3 It has been reported that the number of unhelmeted bike riders in crashes in Florida has risen in the years since wearing a helmet was made non-compulsory. In 1999 when it was compulsory, 22 unhelmeted riders died. By 2004 the number of fatalities had risen to 250. What was the percentage increase in unhelmeted deaths in that time?
- 4 Speed records for bicycles have continued to be broken. It was 223.174 km/h in 1973, 245.08 km/h in 1985 and 268.6 km/h in 1995. What was the percentage increase in speed:
 - a between 1973 and 1985?
 - b between 1985 and 1995?
- 5
 - a The first powered flight by Orville Wright, a bicycle mechanic, was recorded in 1903. He flew 39 metres in 12 seconds at about 3 metres above the ground. How fast did he fly in metres per second?
 - b In 1899, 'Mile-a-Minute Murphy' set a speed record on his bicycle of 1 mile (1.609344 km) in 57.75 seconds. How fast is that in metres per second?
- 6 Lilli's bicycle wheel has a circumference of 223 cm. How fast was Lilli travelling if her wheel made 123 revolutions every minute:
 - a in centimetres per minute?
 - b in kilometres per hour?





- 7 a Brigitte's bike wheels each have a diameter of 70 cm. If the circumference, C , of a circle of radius r is given by $C=2\pi r$, what is the circumference of the wheels in centimetres?
- b Each wheel has 36 spokes. What is the distance around the tyre between spokes?

- 8 a The front wheel of a tricycle has a 20 cm radius. What is the circumference of the wheel in centimetres?
- b Totty pedals through one revolution of the front wheel. How far does she move forward?

- c Totty can pedal at 60 revolutions per minute (1 revolution per second). How far forward does Totty travel each second? How many kilometres per hour is that?

- d Totty's brother can pedal twice as fast. At what speed in kilometres per hour can he travel?

- e Totty's mother wants to ride a tricycle at 27 kilometres per hour. Clearly she can't pedal fast enough on Totty's trike so she will need one with a bigger front wheel. If she wants to pedal at 1 revolution per second, how big does the front wheel need to be so she can ride at 27 kilometres per hour?

$$27 \text{ km/h} = \dots\dots\dots \text{ m/h} = \dots\dots\dots \text{ m/s}$$

So 1 revolution means the trike travels
 metres

and circumference of the front wheel ($2\pi r$)
 = metres

$$\text{So radius of front wheel} = \frac{\dots\dots\dots}{2\pi}$$

= metres

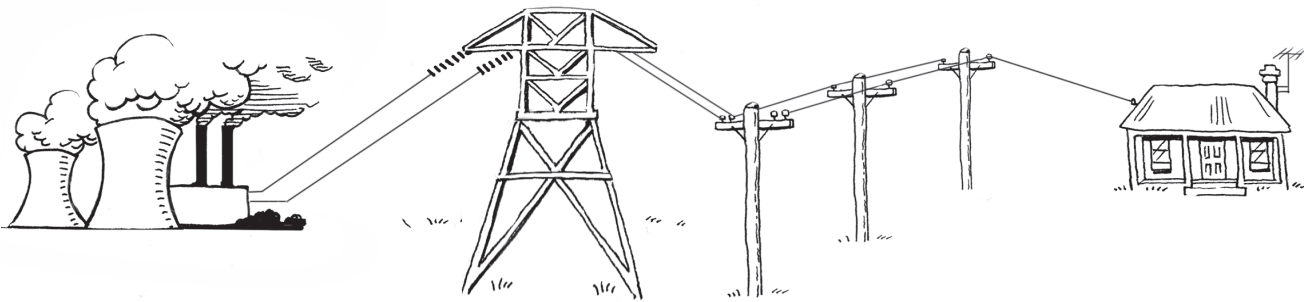


- 9 A chalk mark is drawn on the tyre of a bicycle. Which of the following shows the path the chalk mark makes as the wheel moves forward?



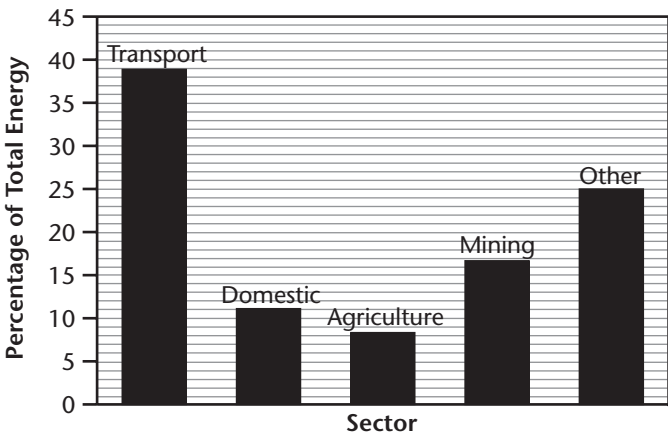


Conserving energy is becoming increasingly important. Understanding how we use energy can help us look after what we have. Greenhouse gas emissions into the atmosphere, including heat-trapping gases such as carbon dioxide, can cause global warming.

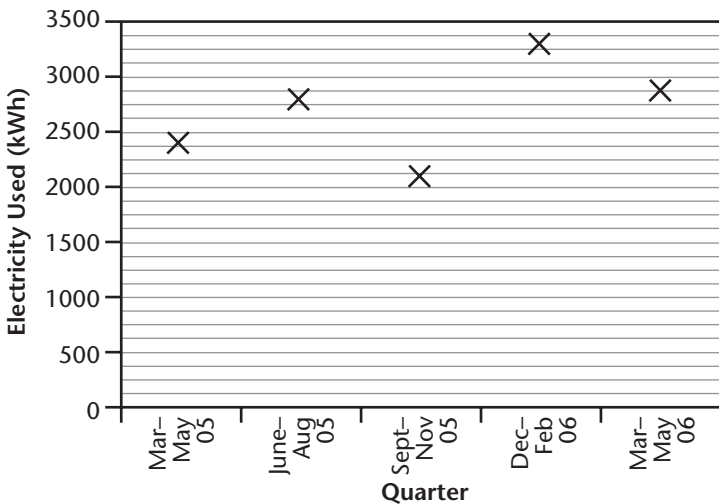


- 1 Appliances left on ‘standby’ draw 13% of the electricity used in Australian homes. My last bill was \$420. How much did it cost me to leave such things as my television, video and computer on standby rather than switching them off?
- 2 It has been suggested that households are responsible for about a third of a state’s emissions, and each year a typical person can pump out 24 tonnes.
 - a What would be a state’s emissions in a year?
 - b Making electricity in power plants was producing 26 tonnes of greenhouse gases each year 10 years ago. If production has increased by 40% in that time, how many tonnes are produced now?

- 3 Use the graph to answer the following questions.
 - a Which sector uses the most energy?
 - b Which sector uses the least energy?
 - c What percentage of energy is used for domestic purposes?
 - d What percentage of energy is used by mining and ‘other’ industries combined?



- 4 The graph shows how much electricity (in kWh) Elly used from March 2005 to May 2006.
 - a Use this graph to complete the table (approximately) on the next page.



The amount of electricity used is measured in kilowatt hours (kWh). If you use a one-bar heater for one hour you will use about 1 kWh.



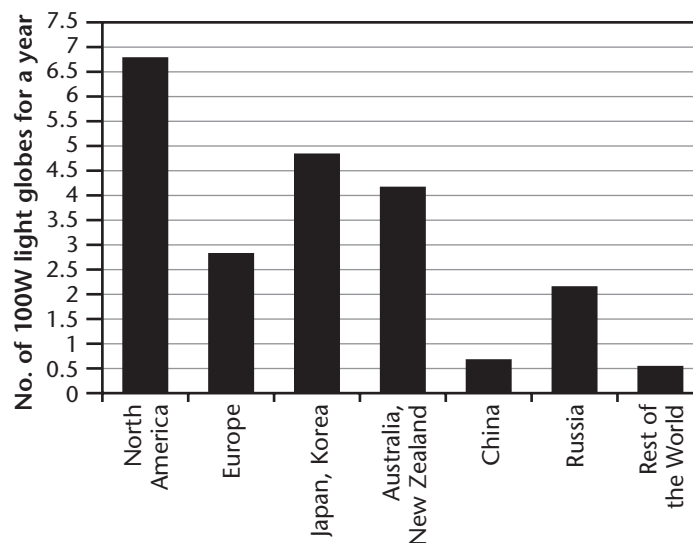
b If electricity costs 12.54 cents per kWh, how much was the largest bill Elly received in the last year?

c What was the total cost of electricity used in the year March 2005–May 2006 inclusive?

- 5** This graph shows the amount of light energy used per person in each of the listed countries in 2005. It is given as the number of 100 watt light globes each person would need to have turned on for a year to use that amount of energy.

Data Reference: www.iea.org

Dates	Total electricity used (kWh)
Mar–May 05	
June–Aug 05	
Sept–Nov 05	
Dec–Feb 05	
Mar–May 06	
Total March 05–May 06 inclusive	



a Which country used the most light energy per person?

b How much light energy did each person in Australia and New Zealand use?

c Read off the values and complete the following table.

Country	100 W globe years
North America	
Europe	
Japan, Korea	
Australia, New Zealand	
China	
Russia	
Rest of the World	
Total light energy used in the world	~22

d Complete this statement:

Each person in the world turned on (number) of 100 watt light globes for a year to give the light energy used.

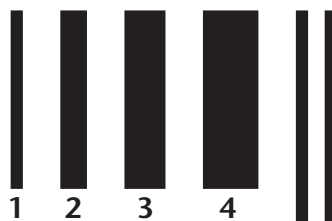


Bar codes were originally devised to help supermarkets speed up their checkouts. Later it was found that they could be linked to inventory and price-control computers. There is some variation in how the barcodes are arrived at, depending on the country. These exercises show examples.

- 1 The barcode shown is from a box of tissues and is made up of two parts: human-readable numbers and a machine-readable barcode.



- a For the numbers, the 9 at the beginning is a code for the country, the next 6 digits are the company that made the tissues, the next five digits indicate what the product is and the last digit is called the 'check digit'. For this box of tissues:
- i the company number is
 - ii the product number is
 - iii the check digit is
- b The bars themselves are rather difficult to decode without a special reader. (They are not the numbers underneath!) There are four different widths for the gaps between the black bars and there are four different widths for the black bars themselves.



There are also double-bar dividers at the beginning, middle and end. List the next four black bars in the barcode for the tissue box by checking their width and then completing the following:

(11), 4, 1, 2, 2,

- 2 Barcodes can be a problem when they need to be placed on a very small package. In these cases, bar codes are shortened by leaving out a set of four digits, all zeros. What does the barcode 0 57000 01235 become when it is shortened?



- 3** The check digit lets the scanner work out if the number has been scanned correctly. Ignore the country code at the beginning and the check digit at the end when you complete the following check on the barcode in question 1.

- a** Find the sum of all the even-positioned digits:

$$1 + 0 + 8 + \dots + \dots = \dots$$

- b** Find the sum of all the odd-positioned digits:

$$3 + 0 + 8 + \dots + \dots + \dots = \dots$$

- c** Multiply the number obtained in Part b by 3:

$$\dots \times 3 = \dots$$

- d** Add the numbers obtained in Parts a and c:

$$\dots + \dots = \dots$$

- e** Find the number that needs to be added to the number obtained in Part d to bring it up to the nearest multiple of 10:

.....

This should be the same as the check digit on the end of the barcode.

- f** The barcode looks genuine, but is it?

- 4** Another box of tissues has the following barcode:

9 310004 14197 6

- a** What is the country code?

- b** What is the check digit?

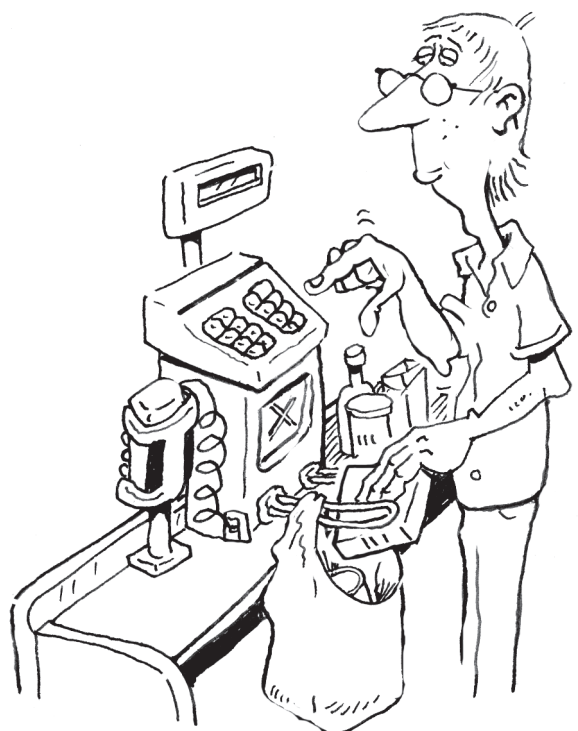
- c** What is the company number?

- d** What is the product number?

- e** Find the sum of the even-positioned digits. Add this to three times the sum of the odd-positioned digits. What is the number that needs to be added to this result to get a multiple of 10?

- f** Is the check digit correct?

- 5** For the barcode 0 57000 01235 x , find x .





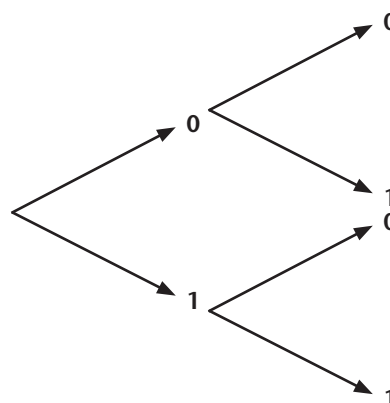
*Bits are **B**inary **di**gi**T**S that are frequently used for storage of information and communication. A bit (symbol **b**) of storage can be either a 1 or a 0, either on or off, either one thing or another. A byte (symbol **B**) is usually eight bits. A nibble (symbol **n**) is 4 bits.*

- 1 The binary number 1101 is 4 bits long because it has 4 digits. Use the table and example to write its decimal equivalent.

e.g.

$2^3 = 8$	$2^2 = 4$	$2^1 = 2$	$2^0 = 1$	Equivalent decimal number
1	0	1	1	$1 \times 8 + 0 \times 4 + 1 \times 2 + 1 \times 1 = 11$
1	1	0	1	

- 2 We wish to find out how many different values can be represented by a 2-bit number. We can do this using a tree diagram as shown. So the first digit can be a 0 or 1 and for each of these, the second digit can be a 0 or 1. Listing them we have 00, 01, 10 and 11 so we can represent four values with a 2-bit number.



- a Use a tree diagram or other method to find out how many values can be represented by a 3-bit number.
 - b List these values.

- 3 A nibble is 4 bits. Find how many values can be represented by a nibble.

- 4 A byte is 8 bits. How many values can a byte represent?

- 5 a How many bytes are there in 3 Megabytes?

- b A nit (natural digit) is 1.44 bits. How many bits in a Meganit?

- 6 My hard drive can store 40 Gigabytes or 40 Gb of information and programs. How many bytes is that?

- 7 The information on the magnetic strip of your credit card is 1288 bits. This is usually made up of 8-bit bytes. How many bytes of information are stored on your credit card?

Decimal Prefixes to Remember

Tera	T	1 000 000 000 000
Giga	G	1 000 000 000
Mega	M	1 000 000
kilo	k	1 000
deci	d	0.1
centi	c	0.01





- 8** Your digital camera was advertised as being 3MB or 3 Megabyte.
- a** The card inside the camera has a capacity of 64 MB. How many pictures can you store on the card before you need to transfer them to your computer?
 - b** A CD holds 600 MB. How many pictures could you store on a CD?
 - c** Your computer can compress each picture so that it only uses 20 kB. How many compressed pictures can you store on:
 - i** the camera card?
 - ii** a CD?
- 9** Stephanie likes to listen to the radio on the Internet and the speed at which she can receive it on her broadband connection is 96 kB (kilobits) per second.
- a** If there are 8 bits (b) per byte (B), at what speed can she receive the radio in kB per second?
 - b** How many kB can Stephanie receive in 1 hour?
 - c** How many kB in a MB?
 - d** How many MB can she receive in 1 hour?
 - e** Her month's allowance on the Internet is 200 MB. How many hours of listening can Stephanie have to use her entire month's allowance.

Pixels are the tiny dots that make up a picture.

The word 'pixel' is short for 'picture element'.

Each pixel has a different brightness of red, green or blue.

1000 x 1000 pixels (bytes) by 3 bytes for colour make the total of 3 million bytes (3MB).

Remember, 1 byte can have 256 values!

- 10** An Internet service provider has the following plans showing monthly charges.

Speed download	Speed upload	Usage 200 MB	Usage 400 MB	Usage 500 MB	Unlimited usage	Usage 20 GB
256 kb/s	64 kb/s	\$29.95	\$35.95	\$45.95	\$59.95	\$79.95
512 kb/s	128 kb/s	\$35.95	\$39.95	\$55.95	\$69.95	\$99.95
1500 kb/s	256 kb/s	\$45.95	\$55.95	\$69.95	\$99.95	\$129.95

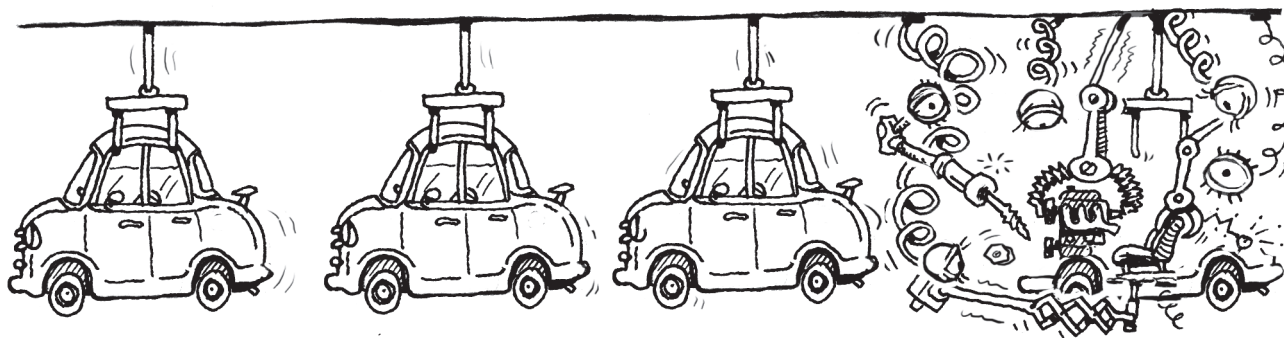
Cost if the plan is exceeded is 15¢ per MB. Usage is the total of uploads and downloads.

Ruth likes to download music and normally uses about 400 MB for that. She usually uploads 100 MB of emails. She prefers to have a speed for download of at least 512 kb/s so she doesn't need to wait too long for her music. Should Ruth choose the 400 MB, 500 MB or the unlimited usage for the cheapest deal?



Probably the most utilised item of modern technology is the motor car. Indeed, it has become a necessity for many of us, without which our daily life would be very different.

- 1
 - a A production company installed 250 robots in its car-making plant. That was said to be one robot for every 14 employees. How many employees were there at the plant?
 - b The company intends to make 140 000 cars each year. How many cars per employee will that be?



- 2 In a university study it was found that the odds of a person being killed or hospitalised doubles for every 5 kph they are travelling above 60 kph.
 - a If the odds of a person being killed or hospitalised in a car crash are 1 to 11 at 60 kph, complete the following table.

Speed (km/h)	Odds of being killed or hospitalised
60	1 : 11
65	2 : 11
70	
75	
80	
85	
90	
95	
100	

In the long run,

$$\text{Pr (being killed or hospitalised)} \approx \frac{\text{number of people killed or hospitalised}}{\text{total number of accidents}}$$

$$\text{Odds of 1 to 11 means Pr (killed or hospitalised)} = \frac{1}{12}$$

$$\text{and Pr (uninjured)} = \frac{11}{12}$$

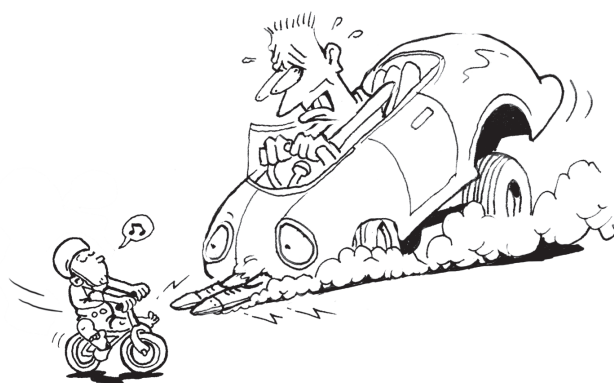
- b What are the odds of you being uninjured if you had a crash at 85 kph?
 - c At what speed would it be a certainty that a person would be killed or hospitalised?
- 3
 - a One of the reasons given in the study for increased risk was a driver's 'reaction time'. The average driver takes 1.5 seconds from the time they see a problem to taking evasive action. During that time the car continues to travel a distance called the 'reaction distance'. Complete the following table for a reaction time of 1.5 seconds using reaction distance = speed $\times$ reaction time.

(Australian Academy of Science)



Speed (km/h)	Speed (m/s)	Reaction distance (m) for reaction time of 1.5 s
60	$\frac{60 \times 1000 \text{ m}}{60 \times 60 \text{ s}} = \frac{100}{6} \text{ m/s}$	$\frac{100}{6} \text{ m/s} \times 1.5 \text{ s} = 25 \text{ m}$
65		
70		
75		
80		
85		
90		
95		
100		

b Two identical cars driven by Mike and Marge are travelling along Finch Street. Mike is travelling at 60 km/h and Marge is travelling at 75 km/h. Mike and Marge are side-by-side as they see little Terry ride his bike out of a driveway into the street in front of them. What is the difference in their reaction distances?



c A driver who is distracted or who has drunk alcohol can have a reaction time of as much as 3 seconds. What is the difference in Mike and Marge's reaction distance if they each have a reaction time of 3 seconds?

4 When the car's brakes are applied, the distance it takes to stop depends on whether the road is dry or wet, whether the car is travelling up or down hill, and how fast the car is travelling. For a car travelling initially (at the beginning) at u metres per second with the rate of deceleration (slowing down) of a metres per second per second, the braking distance d metres can be calculated using the formula:

$$d = \frac{u^2}{2a}$$

Typically the speed drops by about 10 metres per second every second, so $a = 10$. For Mike and Marge's situation in Question 3b, what were their braking distances?

5 Little Terry came running out onto the road chasing a ball. The distance Mike and Marge will travel from the time they see little Terry until they come to a complete stop will be the sums of the reaction distance and the braking distance. Find the distance each driver travels before coming to a complete stop.

6 Little Terry was 41 metres ahead of them when Mike and Marge saw him. Why is Marge more likely to hit little Terry?



Work has different meanings depending on the context. It can be a place of employment, a piece of fine art or handwork like knitting. Work is done on an object by a force to produce energy, to change its position, to lift it up, to change its velocity, and heat it up and so on. Power is how quickly that work is done. Engines do work for us and so does gravity.

- 1** The students in 9Z have decided to knit a blanket and send it to their favourite charity. The blanket will be made up of knitted squares 10 cm by 10 cm. The whole blanket will be 1.8 m wide and 2.4 m long.

a How many squares does the class need to knit?

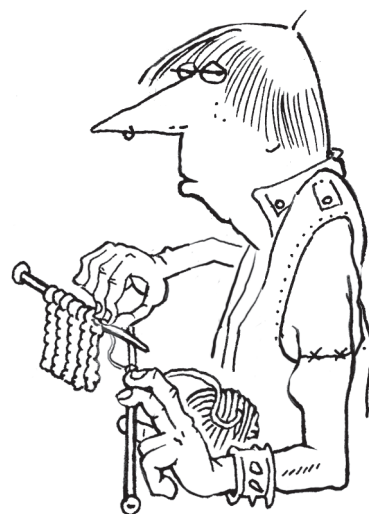
The class is capable of producing 25 squares each week in term time unless they have exams when their production falls by 2 squares per week.

b In a 12-week term of which two weeks are for exams, how many squares can they knit in total?

During the following two week of holidays, the class knitted a total of 60 squares.

c How many squares will they need to knit to finish the blanket?

d At that time, approximately how many weeks should they expect to take to finish the blanket if sewing the squares together will take a week?

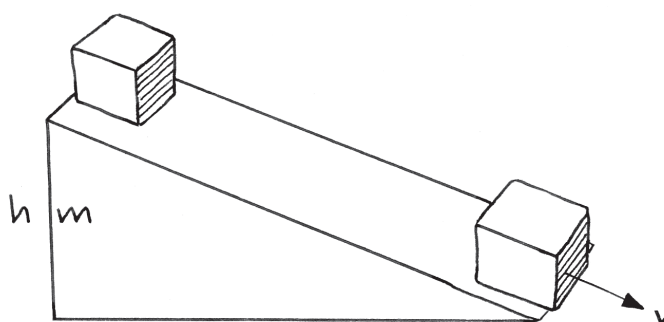


- 2** A block of wood placed at the top of a ramp of height h m starts sliding down. When it reaches the bottom of the ramp it is going at a speed of v metres per second. The slope is perfectly slippery. The speed at the bottom depends only on the height of the slope and can be calculated using the formula:

$$v = \sqrt{19.6h}$$

a Complete this sentence: the higher the ramp, the the block travels at the bottom.

b Use this formula to find the speed of the block at the bottom in metres per second if the ramp is 4 metres high.



The following table shows the speed of the block if it starts at different heights of the ramp.

Height (m)	0.00	0.50	1.00	1.50	2.00	2.50	3.00	3.50	4.00	4.50	5.00	5.50	6.00
Speed (m/s)	0.00	3.13	4.43	5.42	6.26	7.00	7.67	8.28	8.85	9.39	9.90	10.38	10.84



- c Use the table to find the speed of the block at the bottom if the ramp is 2 metres high.
- d Use the table to find how high the ramp needs to be so that the speed at the bottom is 10 m/s.
- e The energy the block has at the bottom of the ramp is given by $E = \frac{1}{2}mv^2$,

where m is the mass of the block in kilograms, v is the velocity in m/s and the unit of energy is joules. How much energy does a 2 kg block sliding down a 3 m high ramp have at the bottom?

- 3 When a skydiver jumps out of a plane, he falls faster and faster under gravity. But then the air resistance starts to work against this until eventually the air resistance balances his body weight. At this point, he stops accelerating until he opens his parachute.

- a What is the terminal velocity in kilometres per hour of a skydiver with the parachute closed?

Idea: $2 \times \text{number of days in September} + 3 \times \text{number of days in March} + \text{number of days in 7 weeks} - 2$

- b What is the terminal velocity in kilometres per hour of a skydiver with arms and legs outstretched? Do you expect the answer to be more or less than in Part a? Why?

Idea: $3 \times \text{number of days in June} + 2 \times \text{number of days in January} + \text{number of days in 6 weeks} - 2$

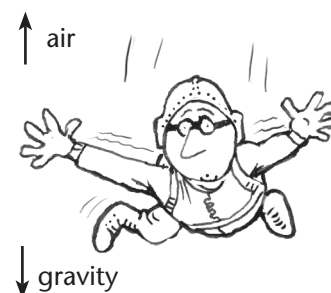
- c The work done by gravity has given the skydiver energy by making him speed up. The amount of energy can be found from knowing how fast he is going and using the formula:

$$E = \frac{1}{2}mv^2$$

where m is the mass of the skydiver in kilograms, v is the velocity in m/s and the unit of energy is joules.

Use a calculator to find the energy of the 72 kg skydiver:

- i with parachute closed and velocity 55 m/s
- ii with arms and legs outstretched and velocity 53 m/s.



- 4 In modern units 1 hp = 746 watts so it could be said that the typical horse described on the right could generate 746 watts of electric power to run a heater or any number of light globes!

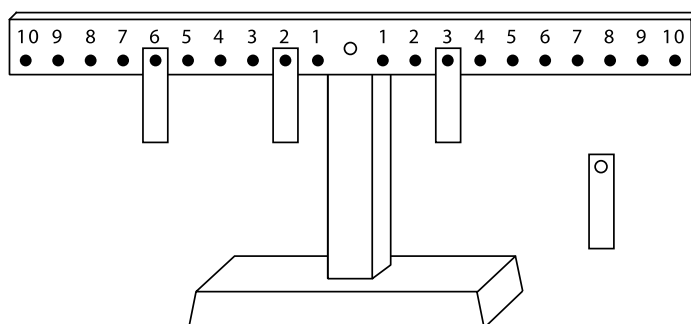
- a How many watts of power could you produce if you can generate about one-tenth of a horsepower?
- b Approximately how many 25 watt globes could you run from one-tenth of a horsepower?

One horsepower (hp) was described by James Watt to help him sell the steam engine he had invented. He thought he should compare the power his engine could produce with what a horse could do. He thought a typical horse could pull a load of 330 pounds (about 150 kg) a distance of 100 feet (about 30 m) in one minute. So this became his definition of the horsepower (1 hp).



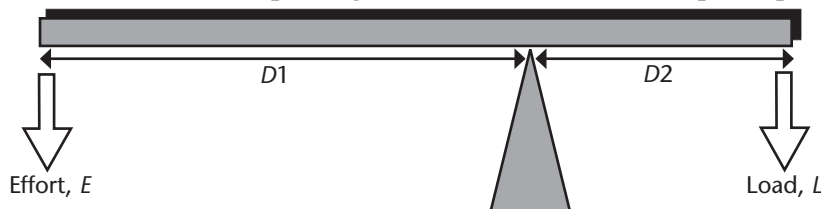
A machine is anything that moves energy from one place to another or changes it from one form to another. So a lever and a pulley, a car and a bicycle are all machines.

- 1 A school buys a 'number equaliser balance' to teach Arithmetic. Equal weights are hung from the numbered rod until it is balanced. In the picture there is one extra weight to be hung.



- Where should the extra weight be hung so that the number equaliser balances? What Arithmetic equation is the balance showing?
- Where would the weights need to be hung to show that $5 + 4 = 6 + 3$?
- How could you show that $10 = 8 + 2$?

The simplest machine is a lever comprising a bar and a fulcrum or pivot point. The load

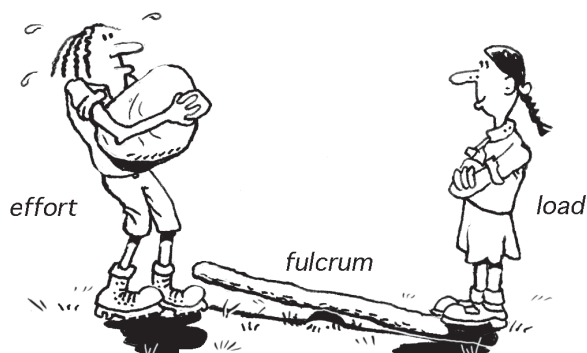


is put on one end of the bar and you apply your effort at the other end. This allows you to move very heavy loads.

The formula for a simple lever to be balanced is: $E \times D1 = L \times D2$

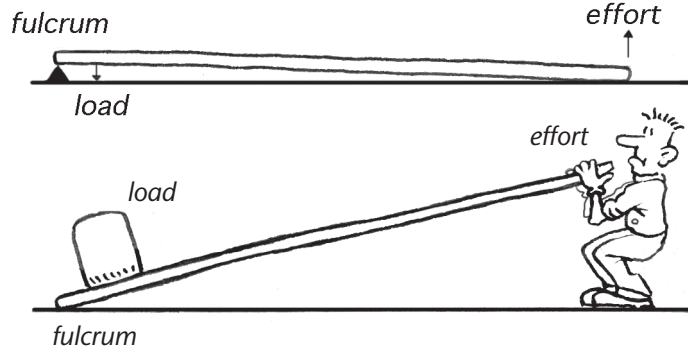
where E is the effort, $D1$ the length of the bar from the effort to the fulcrum, L the load and $D2$ the length of the bar from the load to the fulcrum. The formula works no matter where the fulcrum is.

- 2
- If the student standing on the end of the 2 m long seesaw weighs 80 kg and the rock is being placed on the other end, 1.5 m from the fulcrum:
 - how far is the effort from the fulcrum?
 - how far is the load from the fulcrum?
 - Use the lever formula given above to find how heavy the rock must be so that it:
 - just balances the student.
 - doesn't lift the student off the ground.
 - shoots the student up in the air.





- 3** A crowbar can be used to force open a door or window or to move a load. The fulcrum is fixed at one end and the bar is held at the other. The distance from the effort to the fulcrum is the total length of the crowbar.



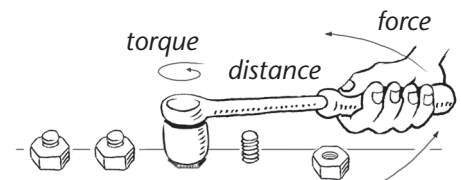
The man moving the rock has a crowbar 1.5 m long and he can apply an effort of 30 kg to one end. The load is 10 cm from the fulcrum.

Use the lever formula to find how heavy a load the man can lift with the crowbar.

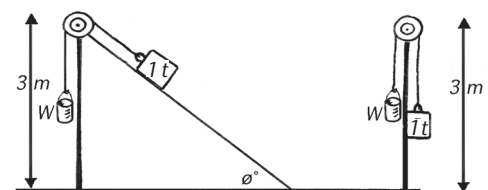
- 4** Increasing the length of the handle of a socket wrench allows you to apply more turning force (torque) to undo or tighten a bolt for example.

Torque = force $\times$ distance

You have a wrench with a 20 cm handle to which you can apply a 24 kg force to the end. If you lengthened the wrench to 60 cm, how much force would you need to use to get the same torque?



- 5** An inclined plane is a simple machine since you can use it to move a weight up more easily than just lifting it straight up. A block of concrete weighing 1 tonne is to be lifted up a height of 3 metres using a pulley and weight W as shown. Of course, the amount of weight needed to lift the concrete straight up must be greater than the weight of the concrete. If the concrete is to be lifted up the slope, the amount of weight needed will depend on the angle θ° and must be greater than $\sin \theta^\circ$ times the weight of the concrete.



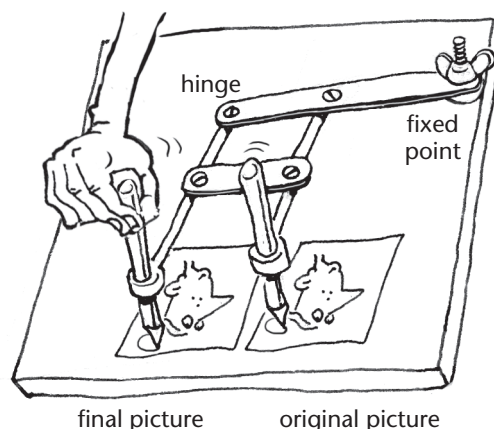
θ°	10°	20°	30°	40°	50°	60°	70°	80°	90°
$\sin \theta^\circ$	0.174	0.342	0.500	0.643	0.766	0.866	0.940	0.985	1.000

- How much weight is needed to lift the concrete straight up?
- How much weight is needed to lift the concrete using a slope of
 - 30°
 - 60°
 - 10°?
- How much weight is needed to lift a piano weighing 450 kg straight up?
- How much weight is needed to lift the piano using a slope of
 - 30°
 - 60°
 - 10°?



Instruments have been invented to help draw and calculate. Many of these have been superseded by modern electronically controlled machines and computers.

- 1 A pantograph is a very old drawing instrument (invented in 1630) used to make scaled copies of pictures. As you move the centre pointer around the original picture, the pencil on the left-hand side automatically draws the copy. The right-hand pointer (the anchor point) is kept fixed. The size of the copied picture can be adjusted by changing the positions of the hinges.

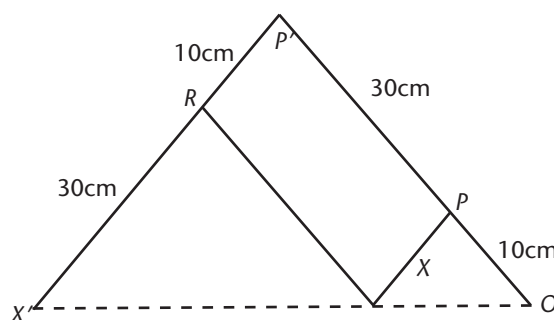
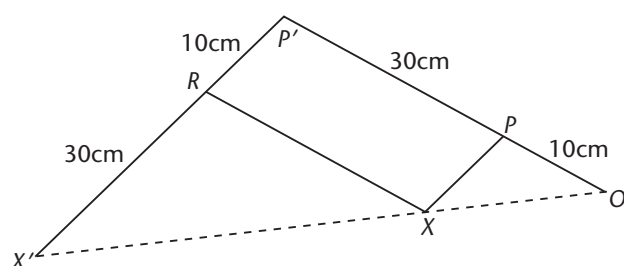
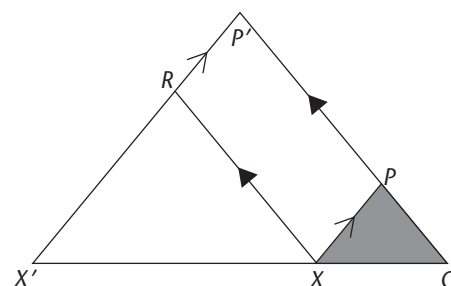


In the diagrams below, the fixed point is at O and the original picture at X . The pencil is at X' . The arms OP' and $X'P'$ are of equal length. Because $X'P'$ is always parallel to XP , the triangles $P'X'O$ and PXO are similar and so the lengths of their sides are in the same ratio.

Comparing similar triangles, $\frac{OX'}{OX} = \frac{OP'}{OP} = \frac{X'P'}{XP}$.

- a The following pantograph (shown in two different positions) has the dimensions:

$OP = 10$ cm, $PP' = 30$ cm, $X'R = 30$ cm, $RP' = 10$ cm. What are the lengths of XP and XR ?



- b Find the ratios of $\frac{OP'}{OP}$ and $\frac{X'P'}{XP}$ and hence $\frac{OX'}{OX}$. What is the magnifying factor of the pantograph?
- c Use ratios to complete the following table to find the length OX' from the fixed point to the pencil for different distances from the fixed point to the original picture drag point.

OX (cm)	2	4	6	8	10
OX' (cm)					

- d The picture to be copied with this pantograph is 6 cm wide and 4 cm high. What is the area of the original picture? What will be the dimensions of the copied picture? What will be the area of the copied picture? What is the magnifying factor for the area?
- e The hinges at P and R are moved so that they are halfway along the arms OP' and $X'P'$. What is the magnifying factor of the pantograph in this arrangement?



f With this new arrangement, the pencil is swapped to where the original picture had been and the picture put where the pencil had been. What is the magnifying factor now? A picture to be copied is 6 cm wide. How wide will this copied picture be?

2 Pantographs can be used on trams or trains to connect the motor to the overhead electric wires.

a Convert 50 km h^{-1} into m s^{-1} by completing the following:

$$\frac{50 \text{ km}}{1 \text{ hr}} = \frac{\dots \text{m}}{\dots \text{min}} = \frac{\dots \text{m}}{\dots \text{sec}} = \dots \text{ms}^{-1}$$

The formula $v^2 = u^2 + 2as$ where $u \text{ m s}^{-1}$ is the starting velocity and $s \text{ m}$ is the distance to reach the final velocity $v \text{ m s}^{-1}$ can be used to find the tram's acceleration a in m s^{-2} .

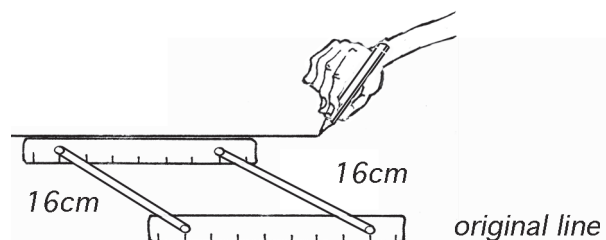
b Use the formula to find in m s^{-2} the acceleration of a tram that travels from 0 to 50 km h^{-1} in 50 m.

c If the acceleration due to gravity, g , is 9.8 ms^{-2} , find how many g s of acceleration that is.

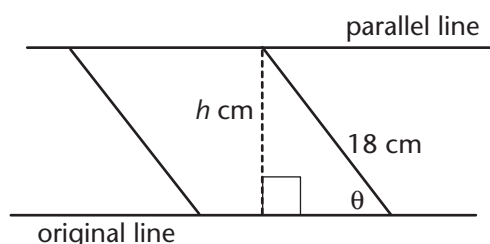
3 A parallel ruler is an instrument for drawing parallel lines. As you move one or other of the rulers, their edges always stay parallel. A given parallel ruler has the dimensions shown.

a What is the maximum distance you can have between the original line and the parallel line you draw if the rulers are 1 cm wide?

b What is the minimum distance you can have between the original line and the parallel line you draw?



The distance, $h \text{ cm}$, between the parallel lines depends on the angle θ and can be calculated using $h = 18 \sin \theta$ from the following diagram:



and the table of values of $\sin \theta$:

θ°	10	20	30	40	50	60	70	80	90	100	110	120
$\sin \theta^\circ$	0.17	0.34	0.50	0.64	0.77	0.87	0.94	0.98	1.00	0.98	0.94	0.87

c Find the distance between the parallel lines when the angle θ is

i 90°

ii 60°

iii 20°

iv 120°

d Explain your answer to Part c iv above.



1 Technology facts and figures

- 1 **a** 20 000 000 **b** 400 000 000
c 1 080 000 000
- 2 42% **3** 199 000 000
- 4 **a** 186 737 **b** 33 460
- 5 **a** 220 000 000 tonnes **b** 24 200 000
- 6 \$9.60; \$48
- 7 **a** 510 740 956 **b** 2 174 291 000
- 8 **a** 204 000 000 **b** 62 769 231 **c** 23 314 286
d 204 000 000 **e** 6 276 923 **f** 13 600 000
- 9 20
- 10 **a** Iceland **b** Tajikistan **c** 32.8 % **d** 0.5%
- 11 **a** jet plane, drummer in a band **b** pneumatic drill, jet plane, grinding machine, paper-making machine, road grader, disco, drummer in a band, USA factory limit **c** No, not if used a lot.

2 Binary numbers and computer switches

- 1 **a** thirty-two point five; $3 \times 10^1 + 2 \times 10^0 + 5 \times 10^{-1}$
b sixty-one point zero; $6 \times 10^1 + 1 \times 10^0 + 0 \times 10^{-1}$
c three hundred and ten point one;
 $3 \times 10^2 + 1 \times 10^1 + 0 \times 10^0 + 1 \times 10^{-1}$
d point three; 3×10^{-1} **e** point zero; 0×10^{-1}
- 2 18 403.9
- 3 $10^2 = 100$, $10^1 = 10$, $10^0 = 1$, $10^{-1} = 1/10$
- 4 $2^2 = 4$, $2^1 = 2$, $2^0 = 1$
- 5 **a** 1 two and 1 one; $16 + 0 + 0 + 2 + 1 = 19$
b 0 twos and 1 one; $8 + 4 + 0 + 1 = 13$
c 0 twos and 1 one; $4 + 0 + 1 = 5$
d 1 two and 1 one; $2 + 1 = 3$
- 6 1001, 1010, 1011, 1100, 1101, 1110, 1111, 10000
- 7 **a** 10 **b** 1100
- 8 **a** A **b** C **c** J **d** P

3 Strength and beauty

- 1 **a** hexagon **b** 60°
- 2 **a** 1 unit³ **b** 8 **c** $\sqrt{2} \approx 1.414$ unit
d $\sqrt{3} \approx 1.732$ unit **e** 6
- 3 **a** 9 **b** $\sqrt{3} \approx 1.732$ unit **c** $\frac{1}{2}\sqrt{3} \approx 0.866$ unit
d 8
- 4 **a** 14 **b** $\sqrt{2} \approx 1.414$ unit **c** $\frac{1}{2}\sqrt{2} \approx 0.707$ unit
- 5 **a** 10 **b** $\sqrt{2} \approx 1.414$ unit **c** $\frac{1}{2}\sqrt{2} \approx 0.707$ unit
d $\sqrt{5} \approx 2.236$ unit **e** 2 unit³
- 6 **a** 14 **b** 1 unit **c** ~ 2.598 unit²
d ~ 5.196 unit³
- 7 **a** 35.8° ; 90° **b** 90° ; 45° **c** 90° ; 60° **d** 70.6°

4 Time passes

- 1 **a** 360° **b** 1080° **c** 810°
- 2 **a** 08:45 **b** 20:45
- 3 Cathy; 55 min **4** 3 o'clock **5** 16:39
- 6 **a** 60° **b** 210° or 150°
- 7 30°
- 8 **a** 00:00 and 11:11 **b** i 0, 1; 2 ii 0, 1, 2, ..., 9; 10 iii 0, 1, 2, 3, 4, 5; 6 iv 0, 1, 2, ..., 9; 10 v $1200 (2 \times 10 \times 6 \times 10)$
c $1200 - 2 = 1198$

5 Calculator words

1 **a, b**

Number	0	1	2	3	4	5	6	7	8	9
Letter (if any)	O	I		E	h	S	g	L	B	

- c** 77345 **d** Teacher to check; e.g. BILL, SHE, BLISS, go... **e** HI **f** LOLLIES **g** Teacher to check; e.g. $7000 + 718 = 7718$ (BILL)
- 2 **a** SHE **b** i 3:37 ii 3:04 **c** i LEE ii HOE
- 3 **a** OHIO **b** i 01:37 ii 04:10 iii 00:51
c i LEIO ii OIHO iii ISOO
- 4 **a** No; shoe would be 3045 and the first digit of the time can only be a 0 or 1, not a 3
b Yes; silo would be 0715
- 5 **a** 0-9 **b** 0, 1, 2 **c** O, I, E, H, S, G, L, B **d** O, I
e No answers given but note that the third digit (second letter) is restricted to 0-5; e.g. LILO, GOGO
- 6 X

6 How significant?

- 1 **a** 2 **b** 3 **c** 2 **d** 4 **e** 5
- 2 **a** 4850 **b** 72.8
- 3 5.0
- 4 **a** 2.48 **b** 2.5
- 5 **a** 0.03 **b** 0.03
- 6 **a** i 2.5 ii 110 **b** 275 or 280 to 2 significant figures
- 7 **a** i 0.33 ii 0.060 **b** 5.5 to 2 significant figures
- 8 3 (down to cm); 2.43 m
- 9 **a** 125.485 g **b** i 3 ii 4 **c** sugar
d 125 g to 3 significant figures
- 10 4.0°C to 2 significant figures since 32 has 2 significant figures.
- 11 7

7 Google

- 1 **a** Archie; 1990 **b** 1995 **c** 1998
- 2 9 million
- 3 3 650 000 000
- 4 **a** 25% **b** 66.67%
- 5 172 800 000
- 6 2 000 000 000 000
- 7 **a** 3 100 000 000 **b** 5898 years
- 8 **a** 800 000 GB **b** 800 000 000 000 000 B
- 9 240 000 000
- 10 **a** \$5 **b** \$157 680 000
- 11 **a** 100 **b** 10^{100} **c** $10^{10^{100}}$
- 12 97
- 13 **a** \$A 1 670 970 682.06 **b** \$A 2 319 146 132.56
c \$A3 763 506 130.87 **d** \$A 2 092 535 448.81

8 The wrong button!

- 1 **a** 200 **b** 200 **c** 400 **d** 394
- 2 **a** 5 **b** 0.3 **c** 5.3 **d** 5.13
- 3 **a** 70 **b** 0.5 **c** 35 **d** 33.12
- 4 **a** $6 \times 10 = 60$ **b** 64.26
- 5 **a** $60 \div 2 = 30$ **b** 31.5
- 6 **a** $30 \times 9 \div 20 = 13.5$ **b** 13.95
- 7 52



- 8 $4 \times 0.4 + 8 \times 0.6 = 1.6 + 4.8 = 6.4$
 9 $6 \times 2 + 90 \div 20 \times 0.05 > 10$
 10 $6 \times 4000 - 30\,000 < 0$; 841.87; first digit rounding of large numbers can give inaccurate results.
 11 $300 \div 6 = 50$ so the result is not correct.
 12 a 300 000 000 b 300 000 000; Yes
 13 a 276 882 478 257 784 b $\frac{1}{175}$
 14 a 2 b 18 c -4 d $\frac{5}{6}$ e $5^5=3125$ f 5
 15 a i 321 321 ii 842 842 b 539 539
 16 a i 323 232 ii 575 757 b 989 898
 17 a 586+ 905 869 b 375 803 758

9 Cooking with altitude

- 1 100°C
 2 70°C
 3 100.06°C
 4 a down (air pressure gets less)
 b up c i 100°C ii $\sim 90^\circ\text{C}$ d $\sim 94^\circ\text{C}$
 e $\sim 85^\circ\text{C}$ f no; $\sim 93^\circ\text{C}$. Note accuracy of answers depends on clarity of the graph.
 5 a Tea will be weaker b 3.2 min c 263 g plain flour, $\frac{1}{2}$ teaspoon baking powder, 50 g white sugar, 56 g brown sugar, 2 tablespoons water, 215°C , 8 min

10 Mobile phones

- 1 a 8 560 872 b 8 449 580.7
 2 857 833 or $\sim 860\,000$
 3 a 4 366 045 b 1 386 861
 4 40 times
 5 a i 18 cm by 6 cm ii 12 cm by 4 cm
 b i 108 cm^2 ii 48 cm^2 c 144:1; 64:1
 6 a 25 m b 2 seconds c 8.33 m
 7 a SOS b - - . - - c E and T d 4 seconds

11 Inventions and patents

- 1 a 50.43% b 171 962 000
 2 12 000 000 km
 3 743.68 m
 4 a 4% b \$296 000 000
 5 a 2004-05; 91 260 b 1993-94; 62 680
 c 4801 d 6.7% e (6261) 7.4%; greater
 6 a 185 b US c 7 d 71
 e i $\sim 1.95 \times 10^{-7}$ ii $\sim 1.54 \times 10^{-7}$
 iii $\sim 5.85 \times 10^{-7}$ iv $\sim 1.87 \times 10^{-7}$
 f United Kingdom $\sim 5.85 \times 10^{-7}$

12 Schools and technology

- 1 a A b C c $44.63 \div 15 = 2.98\text{ m}$ d 2 m
 2

Width (cm)	1	2	3	4	5	6	7	8
Length (cm)	4	7	10	13	16	19	22	25
Area (cm ²)	4	14	30	52	80	114	154	200

5 cm by 16 cm

- 3 a 27% b 4% c 3 years ago

- 4 a Smaller b 2.5 c 9.4 d 9.4
 e negative; $\frac{2.5-9.2}{8} \approx -0.86$ f $y = -0.86x + 9.4$
 g 1.66; 0.8; -0.06; No, because it assumes that computers continue to be bought at the same rate each year.

13 Nuclear energy and green power

- 1 43 500 kg = 43.5 tonne
 2 a USA b France c 104 d $\sim 30\%$ e 15%
 f 45%
 3 a 6 kWh b \$3000 nuclear; \$2050 coal; \$2350 gas; \$3750 wind; \$2500 solar
 4 a 66 000 000 MWh b i 49 500 000 MWh
 ii 9 240 000 MWh iii 660 000 MWh
 iv 5 280 000 MWh c 2837
 5 a

	Homes	Businesses	Total
Amount of energy sold (MWh)	183 000	298 000	481 000
Number of customers	125 000	6 000	131 000
Amount of energy from biomass (MWh)	106 140	172 840	278 980
Amount of energy from wind (MWh)	65 880	107 280	173 160
Amount of energy from hydro (MWh)	9 150	14 900	24 050
Amount of energy from solar (MWh)	1 830	2 980	4 810

b Yes, 115 290

14 Bicycles

- 1 13 420 000
 2 \$15 000
 3 1036%
 4 a 9.82% b 9.60%
 5 a 3.25 m/s b 27.87 m/s
 6 a 27 429 cm/min b 16.46 km/h
 7 a 219.9 cm b 6.11 cm
 8 a 125.7 cm b 125.7 cm c 125.7 cm; 4.53 km/h
 d 9.05 km/h e 27 000 m/h = 7.5 m/s; 7.5 m; 7.5 m; 1.19 m
 9 C, the path is called a cycloid.

15 How we use energy

- 1 \$54.60
 2 a 72 tonnes b 36.4 tonnes
 3 a transport b agriculture c $\sim 11\%$ d $\sim 42\%$
 4 a Mar-May 05, 2400; June-Aug 05, 2800; Sept-Nov 05, 2100; Dec-Feb 05, 3300; Mar-May 06, 2870; Total March 05-May 06 inclusive, 13 470
 b \$413.82 c \$1689.14
 5 a North America b 4.16 100 W globe years



- c** North America 6.8; Europe 2.8; Japan, Korea 4.8; Australia, New Zealand 4.2; China 0.7; Russia 2.2; Rest of the World 0.5; Total ~22
d ~ 22

16 Bar codes

- 1** **a** i 310088 ii 00191 iii 4 **b** 1, 3, 2, 1
2 0 571235
3 **a** 18 **b** 13 **c** 39 **d** 57 **e** 3 **f** No, check digit is 4 and number obtained was 3.
4 **a** 9 **b** 6 **c** 310004 **d** 14197 **e** 18; $12 \times 3 = 36$; 54; 6 **f** Yes, 6
5 9

17 Bits, bytes and nibbles

- 1** $1 \times 8 + 1 \times 4 + 0 \times 2 + 1 \times 1 = 13$
2 **a** $2 \times 2 \times 2 = 8$ values
b 000, 001, 010, 011, 100, 101, 110, 111
3 16
4 $2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 256$
5 **a** 3 000 000 B **b** 1 440 000 b
6 40 000 000 000 B
7 161
8 **a** 21 **b** 200 **c** i 3200 ii 30 000
9 **a** 12 kB per second **b** 43 200 kB in 1 hour
c 1000 **d** 43.2 MB **e** ~4.6 hours
10 400 MB at 512 download/128 upload kb/s and pay extra 15c for each of the 100 MB extra will save \$1 per month on the 500 MB plan.

18 Driving a car

- 1** **a** 3500 **b** 40
2 **a** 60 - 1 : 11; 65 - 2 : 11; 70 - 3 : 11; 75 - 4 : 11; 80 - 5 : 11; 85 - 6 : 11; 90 - 7 : 11; 95 - 8 : 11; 100 - 9 : 11
b $\frac{6}{12}$ or 50:50 **c** 120 kph
3 **a**

Speed (km/h)	Speed (m/s)	Reaction distance (m) for reaction time of 1.5 s
60	$\frac{100}{6}$ m/s = 16.67 m/s	25 m
65	$\frac{650}{36}$ m/s = 18.06 m/s	27.08 m
70	$\frac{700}{36}$ m/s = 19.44 m/s	29.17 m
75	$\frac{750}{36}$ m/s = 20.83 m/s	31.25 m
80	$\frac{800}{36}$ m/s = 22.22 m/s	33.33 m
85	$\frac{850}{36}$ m/s = 23.61 m/s	35.42 m
90	$\frac{900}{36}$ m/s = 25 m/s	37.5 m
95	$\frac{950}{36}$ m/s = 26.39 m/s	39.58 m
100	$\frac{1000}{36}$ m/s = 27.78 m/s	41.67 m

- b** 6.25 m **c** 12.5 m
4 Mike 13.89 m; Marge 21.70 m
5 Mike 38.89 m; Marge 52.95 m
6 Stopping distance for Marge was greater than 41 m, for Mike it was less than 41 m.

19 Doing work

- 1** **a** 432 **b** $250 + 46 = 296$ **c** 76 **d** ~4 weeks
2 **a** faster **b** 8.85 m/s **c** 6.26 m/s **d** ~5.1 m
e 58.83 joules
3 **a** 200 km per hour **b** 192 km per hour; slower because there is more air resistance
c i 108 900 joules ii 101 124 joules
4 **a** 74.6 W **b** 3

20 Machines

- 1** **a** 5 on the right-hand side; $6 + 2 = 3 + 5$
b 5 and 4 on one side and 6 and 3 on the other
c hang 10 on one side and 8 and 2 on the other.
2 **a** i 1.5 m ii 0.5 m **b** i $26\frac{2}{3}$ kg ii less than $26\frac{2}{3}$ kg iii more than $26\frac{2}{3}$ kg
3 450 kg (nearly half a tonne!)
4 8 kg
5 **a** > 1 tonne **b** i > 0.5 tonne ii > 0.866 tonne iii > 0.174 tonne **c** > 450 kg
d i > 225 kg ii > 389.7 kg iii > 78.3 kg

21 Pantographs and parallel rulers

- 1** **a** 10 cm; 30 cm **b** 4:1; 4:1; 4:1; 4 times
c

OX (cm)	2	4	6	8	10
OX' (cm)	8	16	24	32	40

- d** 24 cm²; 24 cm by 16 cm; 384 cm²; 16 times
e 2:1 f $\frac{1}{2}$; 3 cm
2 **a** $\frac{50 \text{ km}}{1 \text{ hr}} = \frac{50000 \text{ m}}{60 \text{ min}} = \frac{50000 \text{ m}}{3600 \text{ sec}} = 13.89 \text{ ms}^{-1}$
b 1.93 m s⁻² **c** 0.197 g
3 **a** 18 cm **b** 2 cm **c** i 18 cm ii 15.66 cm iii 6.12 cm iv 15.66 cm
d Equivalent to result for 60°; $\sin 120^\circ = \sin 60^\circ$

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